

Pigeonhole Principle, Designs, and Extension Elimination

A sequence of lemmas, proofs, and obstructions

Toward superpolynomial fixed-depth

$AC^0[p]$ -Frege lower bounds

What this document contains

The post-upload research developed in the conversation: corrected Koszul vanishing, generic survival, exact nonlinear liftings, joint-moment constructions, polynomial-calculus elimination, batch decompositions, and explicit limits of the proposed methods. Each substantive mathematical result is stated with its hypotheses and a proof or an explicitly identified imported premise. Superseded claims and unsuccessful universal targets are retained as such, not silently promoted to theorems.

Working proof manuscript, not a certified lower bound

The payoff theorem is not established. The main unresolved task is an affordable, refutation-sensitive elimination of the full proof-derived extension system. Computational records support specified examples; they do not certify the general arguments. No claim of priority or independent verification is made.

Compiled from the research conversation

10 September 2026

How to read this record

This is a consolidation of the work after the instruction to stop reading the uploaded research-state PDF and work on the problem directly. It is not a new proof of every assertion in that earlier PDF, nor a verbatim transcript. The uploaded state is listed as a historical starting point; its unrelated claims about functional PHP, circuit measures, and earlier experiments are not silently adopted here.

Proof status. An ordinary numbered lemma below is a working derivation with its proof included. A result using a literature lower bound is explicitly conditional on that imported input. “Open target” means no proof has been obtained. A counterexample applies only to the method or quantifiers named in its statement. Formalization, independent checking, and a complete encoding audit remain outstanding.

Consolidation policy. Repeated results are stated once and later specializations are proved as corollaries. Stronger bounds do not erase earlier constructions with different quantifiers: preserving an arbitrary base functional is different from proving existence of some augmented functional. The original joint-variable degree of each extension axiom is retained throughout. Mathematical scope qualifications made during the conversation are preserved; adding an explicit hypothesis or clarifying a convention is identified in the relevant statement.

External inputs. The published Frege-to-ENS simulation and PHP polynomial-calculus lower bound are inputs, not discoveries claimed here. Their full proofs are not reproduced. Their exact use and the encoding check still needed for the payoff are set out in Sections 1 and 12. Bibliographic records were checked while preparing this manuscript; this was not a new literature survey.

Computations. Eleven previously generated check archives were inspected for their recorded scope and results. They were not rerun during compilation. Appendix A distinguishes finite-domain identities, truncated-design checks, and primitive proof-line checks. Counts are archived reports, not new measurements. A separate source bundle contains the $\text{\LaTeX}$ sources, those archives, and the build log.

Two routes, not one. Sections 2–3 concern a non-Boolean homogeneous ring. The extension-elimination route uses a Boolean algebraic PHP system and, ultimately, PC consequences. Results in the first ring do not automatically pass to the second. The late elimination route can bypass the original generic-linear Lemma B altogether.

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1 Notation, proof systems, and the precise goal

Fix a prime p when working with finite-field extensions; write $k = \mathbb{F}_p$. The early homogeneous-ring results hold over any field. Let n be the number of holes and $m = n + 1$ the number of pigeons. Put

$$\rho_i = \sum_{j=1}^n x_{ij}, \quad \mathcal{F}_n = \{\rho_i - 1\}_i \cup \{x_{ij}x_{i'j} : i \neq i'\} \cup \{x_{ij}^2 - x_{ij}\}_{i,j}.$$

This is the *Boolean linear-row PHP system*. No products from two different columns of one row are set to zero. A Boolean row with $p + 1$ occupied cells can have sum 1 in $\mathbb{F}_p$; hence $\rho_i = 1$ does not mean “exactly one.” The standard nonempty-row clause encoding, this linear-row encoding, and functional PHP are not identified without a proof transformation.

Definition 1 (Bounded consequences and designs). For a finite polynomial system $\mathcal{F} \subseteq k[y]$, set

$$\mathcal{I}_D(\mathcal{F}) = \text{span}_k\{qf : f \in \mathcal{F}, \deg(qf) \leq D\}.$$

A degree- D design is a linear map $\lambda : k[y]_{\leq D} \rightarrow k$ with $\lambda(1) = 1$ and $\lambda(\mathcal{I}_D(\mathcal{F})) = 0$.

A polynomial-calculus derivation uses axioms in $\mathcal{F}$, field-linear combinations of earlier lines, and multiplication of an earlier line by one variable. Its degree is the largest actual total degree of a line. Write $\mathcal{C}_D(\mathcal{F})$ for the vector space of polynomials obtainable by such derivations of degree at most D . A *PC-annihilating functional* additionally satisfies $\lambda(\mathcal{C}_D(\mathcal{F})) = 0$.

Zero polynomials may be omitted; their degree causes no constraint. A Nullstellensatz refutation is an identity $1 = \sum_f q_f f$ with degree $\max_f \deg(q_f f)$. Every degree- D NS refutation is a degree- D PC refutation, but $\mathcal{I}_D$ and $\mathcal{C}_D$ need not be equal.

Lemma 2 (Linear separation). A degree- D design exists exactly when $1 \notin \mathcal{I}_D(\mathcal{F})$. A normalized functional annihilating $\mathcal{C}_D(\mathcal{F})$ exists exactly when $1 \notin \mathcal{C}_D(\mathcal{F})$.

Proof. Both spaces are subspaces of the finite-dimensional vector space $k[y]_{\leq D}$. If 1 lies in the subspace, a functional annihilating it cannot send 1 to 1. Otherwise take a basis of the subspace, append 1, extend to a basis of the ambient space, and define the functional to be zero on the first basis vectors and one on 1. Give arbitrary values, for example zero, on the remaining vectors. $\square$

Lemma 3 (Reusing a PC line versus flattening a certificate). If P has a PC derivation through degree b , then QP has one through degree $\max\{b, \deg Q + \deg P\}$. In contrast, from $P \in \mathcal{I}_b(\mathcal{F})$ one obtains only the automatic bound $QP \in \mathcal{I}_{b+\deg Q}(\mathcal{F})$.

Proof. Retain the derivation of P . Expand Q into monomials, multiply the *final* line P successively by the variables in each monomial, and combine the resulting lines. Every new line has degree at most $\deg Q + \deg P$. For the second claim, multiply a bounded-degree axiom-multiple representation of P by Q ; its summands may use the full old certificate bound b , even when $\deg P$ is small. $\square$

Lemma 4 (Finite-domain reduction with degree control). Suppose every old variable y_v has either a Boolean equation $y_v^2 - y_v$ or an $\mathbb{F}_p$ -field equation $y_v^p - y_v$. Let J_{fld} be their ideal. If $F \in J_{\text{fld}}$ and $\deg F \leq B$, then $F \in \mathcal{I}_B(J_{\text{fld}})$, where the right side means bounded multiples of the displayed generators. Also $g^p - g \in J_{\text{fld}}$ for every $g \in \mathbb{F}_p[y]$.

Proof. Divide by the monic univariate generators one variable at a time. Replacing y_v^e by a lower power never increases total degree, and the subtracted multiple of the appropriate generator has degree no larger than the polynomial being reduced. The final remainder is the

unique representative with individual degrees below 2 or p , respectively. Such a representative is determined by its values on the Cartesian product of the Boolean and field domains: this follows successively from univariate interpolation. Thus a polynomial in the ideal has zero remainder and the division gives a degree-bounded representation. On the finite domain, g takes values in $\mathbb{F}_p$, so $g^p - g$ vanishes and has zero remainder as well. $\square$

Lemma 5 (Indicator identities). For $\alpha \in \mathbb{F}_p$, put $\chi_\alpha(t) = 1 - (t - \alpha)^{p-1}$. Then

$$\sum_{\alpha \in \mathbb{F}_p} \chi_\alpha(t) = 1, \quad t = \sum_{\alpha \neq 0} \alpha \chi_\alpha(t),$$

as exact polynomial identities. For $\alpha \neq 0$,

$$\chi_\alpha(t)(1 - \alpha^{-1}t) = \alpha^{-1}(t^p - t), \quad 1 - t^{p-1} = \prod_{\alpha \neq 0} (1 - \alpha^{-1}t).$$

Proof. The first two expressions have degree at most $p - 1$ and agree with 1 and t at all p field values; univariate interpolation proves equality. For the third, write $1 - \alpha^{-1}t = -(t - \alpha)/\alpha$ and use $(t - \alpha)^p - (t - \alpha) = t^p - t$. The last equality follows from the common roots $\mathbb{F}_p^\times$ and the constant term 1. $\square$

Definition 6 (A leveled extension system). A block a with old-variable inputs $g_{a,1}, \dots, g_{a,k_a}$ and accuracy $h_a \geq 1$ introduces fresh variables $r_{a,u,j}$ and every companion

$$E_{a,i} = g_{a,i} \prod_{u=1}^{h_a} \left(1 - \sum_j r_{a,u,j} g_{a,j} \right).$$

Inputs use only original variables and variables from earlier levels. Distinct blocks in a level have disjoint fresh variables. Include $\mathcal{R} = \{r^p - r\}$ for all new variables. If $\delta_a = \max_j \deg g_{a,j}$ and $g_{a,i} \neq 0$, its original degree is

$$e_{a,i} = \deg g_{a,i} + h_a(\delta_a + 1).$$

An axiom is active at target degree D if its original degree is at most D .

The methods also allow different block accuracies h_a ; when the published simulation is invoked, its common accuracy h is retained. The syntax is the ENS definition in [2, Definition 2.1]. The degree formula follows because the highest-degree terms involving the distinct fresh variable groups cannot all cancel in the polynomial ring.

Lemma 7 (Pointwise soundness and the correct existential quantifier). Any old assignment extends to a scalar assignment to a block's new variables that annihilates all companions and new field equations. This extends through all levels. It does not follow that every scalar assignment works.

Proof. If all inputs vanish, choose all new scalars zero. Otherwise choose $g_j = \alpha \neq 0$, put $r_{1,j} = \alpha^{-1}$, and put all other new variables zero. The common first factor vanishes. Continue level by level. A block with the single input $g = 1$ and all new variables zero instead has companion 1, showing that universal specialization is false. $\square$

1.1 Imported inputs and the payoff

External input E1 (PHP degree). The program uses the published lower bound $\deg_{\text{PC}}(\text{PHP}_n) \geq n/2 + 1$ over every field [3]. Krajíček's exposition records degree- $\lfloor n/2 \rfloor$ designs for the stronger Boolean linear-row system with row exclusions [2, Lemma 4.1]. Deleting those additional axioms preserves designs and cannot lower a degree lower bound. Applications below that use PC rigidity are conditional on this PC input for the stated base encoding; the exact encoding match remains part of the audit.

External input E2 (simulation). The BIKPRS simulation, as stated in [2, Theorem 5.2], takes a fixed-depth Frege refutation with modular counting and polynomially many steps to a leveled ENS/NS refutation with polynomially many companions and degree

$$(O(1) + \log s)(h + 1)^{O(\ell)}.$$

For polynomial s , fixed depth ℓ , and $h = O(\log n)$ this is polylogarithmic. The original paper is [1]. Neither imported theorem is reproved here.

The desired conclusion, for each fixed p , is

$$\forall \ell, K \exists n_0 \forall n \geq n_0 : \text{Size}_{\ell,p}(\text{PHP}_n) > n^K.$$

A sufficient missing theorem would turn each relevant translated degree- D NS refutation into a base PC refutation of degree at most $\lfloor n/2 \rfloor$. The published lower bound would then contradict it. This is an implication to be completed, not a conclusion of this manuscript.

2 The corrected Koszul lemma

For arbitrary row and column counts define

$$S_{r,N} = k[x_{ab} : a \in [r], b \in [N]] / (x_{ab}x_{a'b} : a \neq a'), \quad \sigma_a = \sum_{b=1}^N x_{ab}.$$

Variables may repeat; a monomial cannot use different rows in one column. In the Koszul complex on the row sums, a coefficient variable x_{ab} and the a th exterior generator both have row multidegree ε_a . Internal degree is coefficient degree plus exterior degree.

Lemma 8 (Column factorization). With $C_r = k[y_1, \dots, y_r] / (y_a y_b : a \neq b)$,

$$S_{r,N} \cong C_r^{\otimes N}, \quad \text{Hilb}_{S_{r,N}}(t) = \left(\frac{1 + (r-1)t}{1-t} \right)^N.$$

Proof. All defining relations lie in individual column factors. A basis of one factor is 1 together with y_a^q for $a \in [r]$ and $q \geq 1$. Its Hilbert series is $1 + rt/(1-t)$. Tensor products multiply Hilbert series. $\square$

Obstruction 9 (The originally proposed range $j < n$ is false). For $n = 4$, $m = 5$, the Koszul complex of the row sums on $S_{5,4}$ has nonzero H_1 in internal degree 3, over every field.

Proof. Take row multidegree $\alpha = (1, 1, 1, 0, 0)$. The strand has dimensions

$$0 \longrightarrow k \longrightarrow k^{12} \longrightarrow k^{36} \longrightarrow k^{24} \longrightarrow 0.$$

The degree-zero exterior term has $4 \cdot 3 \cdot 2 = 24$ monomials; the next has $3 \cdot 4 \cdot 3 = 36$; the remaining dimensions are 12 and 1. The first differential from k is nonzero: it sends the exterior product of

the first three row generators to the alternating sum of the three row sums times the corresponding exterior pairs. Thus the next differential has rank at most 11. The kernel at k^{36} has dimension at least $36 - 24 = 12$. Therefore $\dim H_1 \geq 1$. $\square$

Lemma 10 (Multigraded vanishing). Let $\alpha = (\alpha_1, \dots, \alpha_r)$ have all coordinates positive and put $q = \sum_a \alpha_a$. For $i \geq 1$,

$$\boxed{N \geq q + r - i \implies H_i(\sigma_1, \dots, \sigma_r; S_{r,N})_\alpha = 0.}$$

Proof. Separate the last column and put $T = S_{r,N-1}$. There is an exact sequence

$$0 \longrightarrow C_r \longrightarrow \bigoplus_{a=1}^r k[y_a] \longrightarrow k^{r-1} \longrightarrow 0.$$

The first map restricts to the coordinate axes; its image consists of tuples with a common constant term. The last map takes differences of constant terms. Tensor with T . Writing σ'_a for the row sum on the first $N-1$ columns, the row sums act on $T[y_a]$ as $\sigma'_b + \delta_{ab}y_a$, and on each copy of T as σ'_b . The maps respect these actions, so the associated Koszul complexes form a short exact sequence.

In $T[y_a]$, the form $y_a + \sigma'_a$ is monic and hence a nonzerodivisor. Its two-term Koszul complex resolves the quotient T ; tensoring with the remaining free Koszul factors therefore identifies the homology with that of σ'_b , $b \neq a$, on T . The long exact sequence contains

$$\left(H_{i+1}^{r,N-1}(\alpha) \right)^{\oplus(r-1)} \longrightarrow H_i^{r,N}(\alpha) \longrightarrow \bigoplus_a H_i(\sigma'_b : b \neq a; T)_\alpha.$$

Here $H_i^{r,N}(\alpha)$ abbreviates the asserted homology group.

In the right-hand term, row a is absent from the exterior generators and from the differential. Fix its coefficient monomial μ of degree α_a . If $s(\mu)$ columns occur in it, no other row can use those columns, and $s(\mu) \leq \alpha_a$. The corresponding summand is exactly

$$H_i^{r-1, N-1-s(\mu)}(\alpha_a).$$

Induct on r , and for fixed r descend in i . The case $r = 1$ is the Koszul complex of a nonzero linear polynomial in a polynomial ring, and homological indices above r vanish. If $N \geq q + r - i$, then

$$N - 1 \geq q + r - (i + 1),$$

so the left-hand term vanishes by descending induction. For every right-hand summand,

$$N - 1 - s(\mu) \geq (q - \alpha_a) + (r - 1) - i,$$

so induction on r gives its vanishing. Exactness proves the result. Repeated variables and repeated use of a row have not been discarded. $\square$

Lemma 11 (Corrected Lemma (a)). Put $m = n + 1$, $S = S_{m,n}$,

$$\overline{B} = S/(\rho_i - \rho_1 : 2 \leq i \leq m), \quad z = \rho_1, \quad D_0 = \left\lfloor \frac{n+1}{2} \right\rfloor.$$

Over every field, for $0 \leq d \leq D_0$,

$$H_i(\rho_1, \dots, \rho_m; S)_d = 0 \quad (i \geq 1), \quad \dim_k \overline{B}_d = \binom{n}{d} n^d.$$

Multiplication $z : \overline{B}_{d-1} \rightarrow \overline{B}_d$ is injective for $1 \leq d \leq D_0$.

Proof. A row multidegree of total degree d has at most $r \leq d$ active rows. Zero coordinates do not occur in its coefficient monomials or exterior generators. Lemma 10 applies because $n \geq 2d - 1 \geq d + r - i$ for $i \geq 1$. Sum over multidegrees.

Let $\ell_i = \rho_i - \rho_1$. An invertible linear change of generators identifies the full Koszul complex with $K(\ell_2, \dots, \ell_m, z; S)$. Adjoining z is a mapping cone, so its exact sequence includes

$$H_i(\ell; S)_{d-1} \xrightarrow{z} H_i(\ell; S)_d \longrightarrow H_i(\rho; S)_d$$

and

$$H_1(\rho; S)_d \longrightarrow \overline{B}_{d-1} \xrightarrow{z} \overline{B}_d.$$

Starting in internal degree zero, induction gives vanishing of all positive $H_i(\ell; S)_d$ through D_0 . The second sequence gives the asserted injectivity. The mapping-cone statements follow directly by splitting the exterior algebra according to whether its monomials contain the generator for z ; the differential has the two Koszul differentials and multiplication by z as its off-diagonal entry.

By Lemma 8, $\text{Hilb}_S(t) = (1 + nt)^n / (1 - t)^n$. The alternating series of the complex on the n differences is $(1 - t)^n \text{Hilb}_S(t) = (1 + nt)^n$. Positive homology vanishes in the stated range, so the coefficients there equal those of $H_0 = \overline{B}$. This proves the formula and shows $z^d \neq 0$ there. $\square$

Corollary 12 (Weak counting requires no erosion formula). For arbitrary linear forms $L_1, \dots, L_t \in \overline{B}_1$ and $1 \leq d \leq D_0$,

$$\dim(\overline{B}/(L_1, \dots, L_t))_d \geq H_d - tH_{d-1}, \quad H_d = \binom{n}{d} n^d.$$

In particular, the quotient has positive degree- d dimension whenever $t < n$.

Proof. The degree- d part of the ideal is the image of $\overline{B}_{d-1}^{\oplus t} \rightarrow \overline{B}_d$, $(u_s) \mapsto \sum_s L_s u_s$. Its rank is at most tH_{d-1} . Also $H_d/H_{d-1} = n(n - d + 1)/d \geq n$ in the stated range. $\square$

Historical record. The broad range $d < n$ was withdrawn. The conversation also reported $\dim \overline{B}_4 = 3150$ at $n = 5$ over $\mathbb{F}_{1009}$, rather than the extrapolated 3125. That is a recorded computation, not used in any proof here and not rerun for this compilation. The corrected range covers every fixed polylogarithmic degree for sufficiently large n .

3 Generic linear survival and an explicit functional

Continue with the non-Boolean ring $\overline{B}$ of Lemma 11. The original linear version of Lemma B asks for $z^d \notin (L_1, \dots, L_t)\overline{B}$. It is not an annihilator computation and not merely a positive-dimension statement.

Lemma 13 (Affine feasibility and two elementary obstructions). The following are equivalent:

$$z^d \notin \sum_s L_s \overline{B}_{d-1}; \quad \exists \lambda : \overline{B}_d \rightarrow k, \quad \lambda(z^d) = 1, \quad \lambda(L_s u) = 0 \ \forall s, u.$$

The assertion is false for unrestricted linear forms. One form $L_1 = z$ kills z^d . Even when the span of the restrictions does not contain z , the two forms $L_1 = z - x_{11}$ and $L_2 = z - x_{21}$ kill z^2 .

Proof. The equivalence is Lemma 2 in the degree- d vector space with target vector z^d . The first obstruction is immediate. For the second,

$$zL_1 + x_{11}L_2 = z^2 - x_{11}x_{21} = z^2.$$

For $n \geq 2$, comparing coefficients in the other columns modulo row differences shows that no linear combination of these two restrictions equals z . $\square$

Lemma 14 (The degree-two bilinear formulation). At $d = 2$, affine feasibility is equivalent to a symmetric bilinear form $B : \overline{B}_1 \times \overline{B}_1 \rightarrow k$ satisfying

$$B(x_{ij}, x_{i'j}) = 0 \ (i \neq i'), \quad L_s \in \text{rad } B, \quad B(z, z) \neq 0.$$

This applies in characteristic two as well; symmetric does not mean alternating.

Proof. Given a normalized functional, put $B(u, v) = \lambda(uv)$. Conversely a symmetric bilinear form defines a functional on $\text{Sym}^2(\overline{B}_1)$. The displayed column relations are the quadratic defining relations after the linear row differences have already been imposed, so the functional descends to $\overline{B}_2$. The radical condition is exactly annihilation of $L_s \overline{B}_1$. Divide by $B(z, z)$ to normalize. No positivity condition or row-exclusion relation is used. $\square$

Lemma 15 (A regular selected column). In $C_m = k[y_1, \dots, y_m]/(y_i y_j : i \neq j)$, the sum $\sigma = \sum_i y_i$ is a nonzerodivisor and

$$C_m/(\sigma) \cong k \oplus V, \quad \dim V = m - 1, \quad V^2 = 0.$$

Column sums from distinct columns form a regular sequence in their tensor product.

Proof. Write a polynomial uniquely as $a + \sum_i f_i(y_i)$ with $f_i(0) = 0$. Its product with σ is $a \sum_i y_i + \sum_i y_i f_i(y_i)$; distinct coordinate-axis monomials cannot cancel. Modulo σ , multiplying the relation by y_i yields $y_i^2 = 0$, while all mixed products were already zero. There is one linear relation among the m generators. Tensoring over a field preserves injectivity, proving regularity of successive independent factors. $\square$

Lemma 16 (Survival at a maximal-rank specialization). Let $d \geq 1$ and $t + 2d - 1 \leq n$. Impose the first t column sums $\sigma_j = \sum_i x_{ij}$. Then z^d survives in $P = \overline{B}/(\sigma_1, \dots, \sigma_t)$, and

$$H_i(\rho_2 - \rho_1, \dots, \rho_m - \rho_1, \sigma_1, \dots, \sigma_t; S)_j = 0 \quad (i \geq 1, j \leq d).$$

Proof. After quotienting by the regular column sums,

$$S/(\sigma_1, \dots, \sigma_t) \cong S_{m, n-t} \otimes Q, \quad Q = (k \oplus V)^{\otimes t}.$$

Each row sum is $\rho'_i + u_i$ with $u_i \in Q_1$. Filter its Koszul complex by Q -degree, using the descending filtration so the differential preserves it. The associated graded piece of Q -degree a in total degree j is $K(\rho'; S_{m, n-t})_{j-a} \otimes Q_a$. Lemma 10 gives zero positive homology there because $n - t \geq 2d - 1$. The filtration is finite in each internal degree. Successively lifting a leading graded boundary eliminates every positive-degree cycle; equivalently, the finite filtered-complex spectral sequence has no positive homology on its first page. Thus the full row-sum complex over the quotient has zero positive homology through d .

The regular-sequence reduction identifies this with the homology of all row sums and the selected column sums in S . Change the row generators to the differences and z . The same mapping-cone sequence as in Lemma 11 proves both the asserted positive-homology vanishing and injectivity of $z : P_{j-1} \rightarrow P_j$ through d . As $P_0 = k$, $z^d \neq 0$. $\square$

Lemma 17 (Generic survival with a determinant formula). For $t + 2d - 1 \leq n$, there is a polynomial $\Delta(C)$ in the coefficients of t linear forms, nonzero at a specified k -point, such that for $\Delta(C) \neq 0$ a normalized annihilating functional exists. It is given explicitly by a ratio of determinants.

Proof. Use the monomial basis of S_d . Let $A(C)$ have as columns all products of a row difference or one of the t coefficient-dependent linear forms with basis monomials of S_{d-1} . Set $v = [\rho_1^d]$. Specialize the forms to the selected column sums, obtaining C_0 , and put $r = \text{rank } A(C_0)$.

The previous lemma gives $v \notin \text{im } A(C_0)$ and exactness at the first term of the degree- d Koszul complex

$$K_{2,d} \xrightarrow{B(C_0)} K_{1,d} \xrightarrow{A(C_0)} S_d.$$

Consequently, $\text{rank } A(C_0) + \text{rank } B(C_0) = \dim K_{1,d}$. Over $k(C)$, each rank is at least its specialized rank, while $A(C)B(C) = 0$ forces the sum to be at most $\dim K_{1,d}$. Both ranks must stay equal to their specialized values. Thus all $(r + 1)$ -minors of $A(C)$ vanish identically, and $\text{rank } A(C) \leq r$ at every specialization.

Choose r independent column indices J at C_0 and $r + 1$ row indices R such that

$$\Delta(C) = \det[A(C)_{R,J} \mid v_R]$$

is nonzero at C_0 . For $f \in S_d$, define

$$\lambda_C([f]) = \frac{\det[A(C)_{R,J} \mid [f]_R]}{\Delta(C)}.$$

When $\Delta(C) \neq 0$, the selected columns remain independent and span the entire relation image, since its rank is at most r . Replacing the last column by any relation vector gives determinant zero. The expression is linear in f , descends through the row-difference quotient, and sends z^d to 1. $\square$

Corollary 18 (Affine design and generic dimension). For $f = \sum_{e=0}^d f_e$, define

$$\Lambda_C(f) = \lambda_C\left(\sum_{e=0}^d z^{d-e} f_e\right).$$

This is a degree- d design for column exclusion, $\rho_i - 1$, and the chosen linear equations. In the same generic region, the degree- d quotient dimension is the coefficient of u^d in $(1 + nu)^n(1 - u)^t$.

Proof. Homogenization converts a row-axiom multiple to a multiple of $\rho_i - z$ and converts the other required multiples into the relation space annihilated above. Normalization follows by homogenizing 1 to z^d . The dimension statement is the Euler characteristic of the exact low-degree Koszul complex on the $n + t$ forms; maximal rank transfers it from the special point to the same open region. $\square$

Scope. This proves the working generic lower bound $t \geq n - 2d + 1$, not survival for all coefficients and not a finite-field probability estimate. A nonzero polynomial can vanish on a constrained set of $\mathbb{F}_p$ -points. The known good point need not belong to an extension-derived coefficient family. These functionals have not been shown to annihilate Boolean equations.

3.1 The Boolean target and the missing transfer

The homogenized Boolean base ring is

$$A_n = k[x, z]/(\rho_i - z, x_{ij}x_{i'j}, x_{ij}^2 - zx_{ij}) \cong \overline{B}/(x_{ij}^2 - zx_{ij}).$$

It is a further quotient: survival in $\overline{B}$ says nothing automatic about survival in A_n . A linear extension factor $1 - \sum_j \beta_j g_j$ homogenizes to $z - \sum_j \beta_j g_j$ when the g_j are homogeneous linear, not to the latter sum alone. General inputs are nonlinear; companions are products; and the number of factors per block is not the number of blocks.

A fully specified sufficient target is a normalized degree- D design for the entire joint system $\mathcal{F}_n \cup \mathcal{E} \cup \mathcal{R}$, separately for each admissible $\mathcal{E}$. A scalar-specialized design is stronger than necessary. Requiring all multipliers allowed by the *actual post-specialization degree* is stronger still than the original-joint-degree condition used below. The later lifting and elimination results deliberately avoid assuming these stronger requirements.

4 Exact reweighted and shared-conditioning liftings

These constructions change the old moments. Their generic form is

$$\Lambda(P(y, r)) = \frac{\lambda(W(y)P(y, \beta))}{\lambda(W)}, \quad \lambda(W) \neq 0,$$

with a scalar assignment β to the new variables. They use finite-field identities, not approximate satisfaction.

Lemma 19 (One-block reweighted lifting). Let $\mathcal{F} \subseteq \mathbb{F}_p[y]$ contain the appropriate Boolean or field equations. For a block of accuracy h , put $\delta_i = \deg g_i$, $\delta = \max_i \delta_i$, and $e_i = \delta_i + h(\delta + 1)$. Define

$$c(D) = \max(\{0\} \cup \{D - e_i + (p - 1)\delta_i : e_i \leq D, g_i \neq 0\}).$$

Every degree- $(D + c(D))$ design for $\mathcal{F}$ gives a degree- D design for the block-augmented system and its new field equations.

Proof. First ask whether $\lambda(qg_i) = 0$ for every active i and every monomial q of degree at most $D - e_i$. If so, take $W = 1$ and set all new variables to zero. A permitted multiplier $H(y, r)$ specializes to a polynomial of degree at most $D - e_i$, so the companion $E_i(y, 0) = g_i$ is annihilated with all its allowed multiples.

Otherwise choose such i, q with $\lambda(qg_i) \neq 0$. By Lemma 5, some $\alpha \neq 0$ has $\lambda(q\chi_\alpha(g_i)) \neq 0$. Set

$$W = q\chi_\alpha(g_i), \quad \beta_{1,i} = \alpha^{-1}, \quad \beta_{u,j} = 0 \text{ otherwise.}$$

For every companion, the exact identity

$$WE_j(y, \beta) = \alpha^{-1}qg_j(g_i^p - g_i) \in J_{\text{fld}}$$

holds. It persists after multiplication by any polynomial. Also $\deg W \leq D - e_i + (p - 1)\delta_i \leq c(D)$. For every allowed H , the weighted specialized product has degree at most $D + c(D)$, so Lemma 4 lets the base design annihilate it. Old-axiom multiples remain old-axiom multiples within the same budget. New field equations vanish at β , and division by $\lambda(W)$ normalizes the map. $\square$

Lemma 20 (One weight for a level of repairs). For a level of s disjoint blocks, define $c_a(D)$ as in Lemma 19. A degree

$$B = D + \sum_{a=1}^s c_a(D)$$

base design yields a degree- D joint design for the whole level.

Proof. Begin with $W = 1$ and mark no blocks handled. Check every currently unhandled block for a nonzero moment $\lambda(Wqg_{a,i})$ in its allowed multiplier range. If none exists, assign all those blocks zero scalars and stop. Otherwise multiply W by $q\chi_\alpha(g_{a,i})$, choosing $\alpha \neq 0$ so the new value of W under λ remains nonzero; specialize that block by the corresponding nonzero coefficient in its first factor; mark it permanently handled.

Each repair adds at most $c_a(D)$ to the weight degree and handles a new block. For an already handled block, the weight times every specialized companion belongs to J_{fld} . Multiplying the weight later preserves ideal membership. The unhandled blocks are rechecked after every change, so their final zero-specialization conditions are valid for the *final* weight. The remaining checks are those of Lemma 19. $\square$

Corollary 21 (Repair budgets through levels). If $h \geq p - 2$, then $c_a(D) \leq \max\{0, D - h\}$, so one level costs at most $D + s \max\{0, D - h\}$. For levels $1, \dots, \ell$, set backward

$$D_\ell = D, \quad D_{j-1} = D_j + \sum_{a \text{ in level } j} c_a(D_j).$$

A degree- D_0 base design lifts through all levels. In particular, $D_0 \leq D \prod_j (1 + s_j)$ when level j has s_j blocks.

Proof. For an active companion,

$$D - e_i + (p - 1)\delta_i = D - h(\delta + 1) + (p - 2)\delta_i \leq D - h.$$

Apply the level lemma successively, treating prior axioms and their field equations as old axioms. Keep original formal degrees even when the functional evaluates some variables at scalars. The coarse product bound follows from $D_{j-1} \leq (1 + s_j)D_j$. $\square$

Lemma 22 (Shared-feature conditioning). Let J be either the old Boolean/field ideal or the Boolean column-exclusion ideal, in both cases with degree-nonincreasing normal-form reduction. Suppose there are polynomials $f_1, \dots, f_R$ in the original variables such that every input, including later-level inputs, satisfies

$$g_{a,i}(x, r_{<a}) \equiv G_{a,i}(f_1(x), \dots, f_R(x), r_{<a}) \pmod{J}.$$

Put $\kappa = (p - 1) \sum_\nu \deg f_\nu$. A degree- $(D + \kappa)$ base design yields a degree- D design for the entire leveled extension system.

Proof. For $\alpha \in \mathbb{F}_p^R$, define $W_\alpha = \prod_\nu \chi_{\alpha_\nu}(f_\nu)$. These polynomials sum to 1, so some $W = W_\alpha$ has $\lambda(W) \neq 0$. Its degree is at most κ . On the cell selected by W , all common features are scalar constants. Proceed through the levels. Once earlier extension variables have been specialized, every current input has a fixed value on that cell. If all are zero, set the new block to zero. Otherwise use the inverse of one nonzero input in the first factor. Every companion vanishes on the cell. Hence $WE_{a,i}(x, \beta) \in J$ for every block and level.

The quotient by J is the algebra of functions on the corresponding finite domain. In the Boolean column-exclusive case this follows from the independent column-state indicator basis. Thus the pointwise assertion is exactly ideal membership. The assumed degree-nonincreasing reduction gives bounded certificates for the weighted products, whose degrees are at most $D + \kappa$. Old axioms and new field equations are treated as before. No further weight is introduced as levels are traversed. $\square$

Corollary 23 (Selecting the shared cell without exhaustive enumeration). Given access to λ , a suitable W in Lemma 22 can be found using at most pR value queries.

Proof. If a prefix weight W' has nonzero value, then $\sum_{\alpha \in \mathbb{F}_p} \lambda(W' \chi_\alpha(f_\nu)) = \lambda(W') \neq 0$. At least one next value preserves nonzeroness. Choose it and continue. $\square$

Lemma 24 (Conditioning on whole PHP columns). If all original-variable dependence of the extension system lies in the same q columns, then shared conditioning costs only q degrees:

$$B = D + q.$$

There is no restriction on the number of blocks, levels, or nonlinear inputs.

Proof. For a column j , its possible states are empty or occupied by one row. Their indicators are

$$e_{j,0} = 1 - \sum_i x_{ij}, \quad e_{j,i} = x_{ij} \quad (1 \leq i \leq m).$$

They sum to 1, are affine, and are mutually exclusive modulo Booleanity and column exclusion. A product selecting states of q columns has degree q . Such products sum to 1, so one has nonzero design value. On its cell the specified columns are completely fixed. Apply the preceding proof. $\square$

Scope. With the imported degree- $\lfloor n/2 \rfloor$ PHP design, the sufficient shared budget is $D + q \leq \lfloor n/2 \rfloor$ or $D + (p-1) \sum \deg f_\nu \leq \lfloor n/2 \rfloor$. Taking all n columns is always possible as a description, but yields $D + n$, which is unaffordable. No universal cheap common-feature description was proved.

5 Factor packing, hybrid budgets, and a pointwise barrier

Lemma 25 (Packing one block into polynomial substitutions). Let R be the dimension of the polynomial span of a block's input tuple, and let δ bound their degrees. There is a polynomial substitution $r = \beta(y)$ that makes every companion belong to J_{fld} , with

$$\deg \beta_{u,j} \leq L := \delta \left(\left\lceil \frac{(p-1)R}{h} \right\rceil - 1 \right)_+.$$

Proof. Choose a basis $f_1, \dots, f_R$ from the inputs and put $Z = \prod_\nu (1 - f_\nu^{p-1})$. Every $g_i Z$ belongs to J_{fld} , since $f_\nu Z = (f_\nu - f_\nu^p) \prod_{\mu \neq \nu} (1 - f_\mu^{p-1})$ and the g_i are constant-coefficient combinations of the basis.

Factor each $1 - f_\nu^{p-1}$ into the $p-1$ factors $1 - \alpha^{-1} f_\nu$ and distribute all $(p-1)R$ factors into h groups, each of size at most $Q = \lceil (p-1)R/h \rceil$. For a group with ordered factors $1 - \alpha_1^{-1} f_{\nu_1}, \dots, 1 - \alpha_b^{-1} f_{\nu_b}$, define

$$\beta_{u,\nu} = \sum_{\ell: \nu_\ell = \nu} \alpha_\ell^{-1} \prod_{v < \ell} (1 - \alpha_v^{-1} f_{\nu_v}).$$

Telescoping gives

$$1 - \sum_{\nu} \beta_{u,\nu} f_{\nu} = \prod_{\ell=1}^b (1 - \alpha_{\ell}^{-1} f_{\nu_{\ell}}).$$

Set coefficients of unused original inputs to zero. The product across groups is Z , so the companion image is $g_i Z$. Each coefficient has at most $Q - 1$ input factors and therefore degree at most L . If $R = 0$, all companions are zero and every coefficient can be zero. $\square$

Lemma 26 (A whole level costs a maximum, not a sum). For a level of blocks define

$$T_a = \max \left\{ 1, \delta_a \left(\left\lceil \frac{(p-1)R_a}{h_a} \right\rceil - 1 \right)_+ \right\}, \quad T = \max_a T_a.$$

A degree- TD base design yields a moment-preserving degree- D joint design under $\Lambda(P) = \lambda(P(y, \beta(y)))$. Across levels the sufficient budget is $D \prod_j T_j$.

Proof. All substitutions within a level depend only on old variables, so they do not nest inside one another. Substitution sends a degree- D polynomial to degree at most TD . Every companion image is an old-field-ideal polynomial and every new field equation maps to $\beta^p - \beta \in J_{\text{fld}}$. Lemma 4 supplies degree-bounded certificates. An old-axiom multiple remains a multiple of that old axiom. The map preserves normalization and old moments. Compose these substitutions through successive levels to obtain the product bound. $\square$

Corollary 27 (Free packing ranges). Over $\mathbb{F}_2$, linear input spans of dimension at most $2h$ admit $T = 1$. Over $\mathbb{F}_p$, the same holds for linear spans of dimension at most $\lfloor 2h/(p-1) \rfloor$. If all basis inputs are Boolean-valued modulo the old field ideal, replace $(p-1)R$ by R in the factor count. One may also restrict the basis to the span of active inputs, retaining original companion degrees.

Proof. The first two statements make every group contain at most two linear factors, so the substitution coefficients are affine. Boolean-valued inputs have the zero indicator $1 - f$ directly. Inactive companions impose no constraints at this degree; unused coefficients can be set to zero and their field equations still vanish. Active inputs span all companions that must be handled. $\square$

Lemma 28 (Hybrid polynomial packing and scalar repair). Choose an integer $\tau \geq 1$. Pack blocks with $T_a \leq \tau$ and reserve $\mathcal{H}_{\tau} = \{a : T_a > \tau\}$ for repairs. Define

$$c_a^{(\tau)}(D) = \max(\{0\} \cup \{\tau(D - e_{a,i}) + (p-1)\delta_{a,i} : e_{a,i} \leq D\}).$$

A sufficient budget is

$$\min_{\tau \geq 1} \left[\tau D + \sum_{a \in \mathcal{H}_{\tau}} c_a^{(\tau)}(D) \right].$$

It is no worse than either the full repair bound or the full packing bound.

Proof. After the packed substitutions, an allowed multiplier of $E_{a,i}$ has old variable degree at most $\tau(D - e_{a,i})$. Run the repair algorithm on the remaining blocks with that enlarged moment range. Each repaired block adds at most $c_a^{(\tau)}(D)$ to the weight. Already packed companions belong to J_{fld} , so multiplying the weight cannot spoil them. Every final weighted polynomial has degree at most $\tau D + \deg W$. Take $\tau = 1$ to recover a bound no worse than repairing every block, and $\tau = \max_a T_a$ to recover full packing. Between distinct values of T_a , the reserved set is constant and the cost is nondecreasing, so those thresholds and 1 suffice for the optimization. $\square$

Corollary 29 (Packing plus shared conditioning). If one part of a level has substitution degree at most T and the other part is determined by shared features of total indicator cost κ , a sufficient budget is $TD + \kappa$.

Proof. Choose the shared weight and scalar specializations for the second part, and use polynomial substitutions for the first. Packed companions already belong to the old-field ideal, hence remain annihilated after multiplying the weight. The combined weighted substitution has degree at most $TD + \kappa$. $\square$

Obstruction 30 (Sharp pointwise selector degree). For independent field inputs $g_j = y_j$, $1 \leq j \leq R$, any pointwise-valid polynomial substitution of maximum degree L satisfies

$$L \geq \left\lceil \frac{(p-1)R}{h} \right\rceil - 1.$$

The factor construction attains this bound for these inputs. On a Boolean domain, the corresponding argument gives $R \leq h(L+1)$.

Proof. Let $P = \prod_{u=1}^h (1 - \sum_j \beta_{u,j}(y)y_j)$. Pointwise validity requires $y_j P = 0$ for all j . Thus P is 1 at zero and 0 elsewhere on $\mathbb{F}_p^R$. Its unique reduced representative is $\prod_j (1 - y_j^{p-1})$, of degree $(p-1)R$. Reduction does not increase degree, whereas $\deg P \leq h(L+1)$. For Boolean inputs use the reduced representative $\prod_j (1 - y_j)$. The bound for independent field inputs agrees with Lemma 25 at $\delta = 1$. $\square$

5.1 A barrier to annihilating modulo column equations alone

Let

$$A = k[x]/J, \quad J = (x_{ij}^2 - x_{ij}, x_{ij}x_{i'j} : i \neq i').$$

This is the function algebra on all column-exclusive Boolean assignments, without row normalization. For an element of A , $\deg_A$ denotes its smallest representative degree.

Lemma 31 (An empty-row-supported function has full degree). If $0 \neq f \in A$ vanishes whenever row i is nonempty, then $\deg_A f = n$.

Proof. In column j use the affine basis 1 and the state indicators $e_{j,a}$ for all states $a \neq i$, including the empty state. The omitted indicator is $x_{ij} = 1 - \sum_{a \neq i} e_{j,a}$. The change of basis and its inverse are affine, so the degree filtration is exactly the number of nonconstant tensor factors. In the unique tensor expansion of f , restrict column j to state i . Every indicator in that factor vanishes. Since the restricted function is zero, all coefficients of terms omitting column j vanish. Do this for every column. Only terms using all n columns remain. At least one is nonzero, proving both the lower bound and the degree- n upper bound. $\square$

Obstruction 32 (Weighted pointwise packing can require degree above n). For each row i , use the extension inputs $g_{i,j} = x_{ij}$ and write

$$P_i(x, r) = \prod_{u=1}^h \left(1 - \sum_j r_{i,u,j} x_{ij} \right).$$

Suppose $W \neq 0$ in A and substitutions of maximum degree L satisfy $W x_{ij} P_i(x, \beta) = 0$ in A for all i, j . Put $\kappa = \deg W$ and $T = \max\{1, L\}$. Then

$$\kappa + h(L+1) \geq n.$$

For target $D \geq 2h + 1$, the sufficient weighted-substitution budget necessarily obeys

$$\boxed{\kappa + TD \geq n + D - 2h \geq n + 1.}$$

Proof. Let $Q_i = \prod_j (1 - x_{ij})$. Since multiplication by x_{ij} kills $WP_i(x, \beta)$, multiplication by $1 - x_{ij}$ leaves it unchanged. Hence $WP_i = Q_i WP_i$. Also $Q_i P_i = Q_i$, so $WP_i = WQ_i$ in A . Choose an assignment where W is nonzero. Some row is empty because there are $n + 1$ rows and only n occupied-column opportunities. For that row $WQ_i \neq 0$. Lemma 31 gives

$$n = \deg_A(WP_i) \leq \kappa + h(L + 1).$$

If $L \geq 1$, then $\kappa + DL \geq n - h(L + 1) + DL \geq n + D - 2h$ since $D > h$. For $L = 0$, $\kappa + D \geq n - h + D$ is even stronger. Moreover P_i itself has original degree $2h \leq D$, yet its weighted image has normal-form degree n . Thus exact normal-form reduction does not rescue a base functional defined only through degree $n/2$. $\square$

Lemma 33 (Sharpness of the pointwise budget, and a zero-cost escape). If $n \geq 2h$, the pointwise degree bound is attained formally with $\kappa = n - 2h$ and $T = 1$. Nevertheless all of these row blocks lift at degree D with no weight once the base equations $\rho_i - 1$ are allowed.

Proof. For sharpness take $W = \prod_{j=1}^{n-2h} x_{jj}$. For an already selected row, use the factor $1 - x_{ii}$. For any remaining row pair the $2h$ remaining columns. For a pair a, b , set coefficients $\beta_a = 1$, $\beta_b = 1 - x_{ia}$, obtaining $(1 - x_{ia})(1 - x_{ib})$. The product and W annihilate every companion modulo J . This asserts a nonzero formal weight in A , not a nonzero value under every prescribed base functional.

For the zero-cost escape set $r_{i,1,j} = 1$ for all j and all other factor coefficients zero. The companion becomes

$$x_{ij}(1 - \rho_i) = -x_{ij}(\rho_i - 1).$$

A permitted multiplier has degree at most $D - (2h + 1)$, so its specialized base-axiom multiple has degree at most $D - (2h + 1) + 2 \leq D$. Compose any base design with this constant substitution. This uses row equations, not J alone, and therefore does not contradict the obstruction. $\square$

6 Base-aware certificates and structured nonlinear liftings

Lemma 34 (Certificate-based scalar lifting). Let λ be a degree- D design for $\mathcal{F}$. Suppose a scalar specialization β has, for every active extension axiom E of original degree e_E , a certificate

$$E(y, \beta) = \sum_{f \in \mathcal{F}} A_{E,f} f, \quad \deg(A_{E,f} f) \leq e_E.$$

Then $\Lambda(P) = \lambda(P(y, \beta))$ is a degree- D design for the augmented system, including new field equations.

Proof. A permitted cofactor has degree at most $D - e_E$. Multiply the displayed certificate by its specialization; every term then has degree at most D . Old-axiom multiples remain such multiples, constants specialize without degree increase, and new field equations vanish. Normalization is unchanged. $\square$

Unrestricted ideal membership would be useless here, since the base system is unsatisfiable. The certificate degree is the substance of the hypothesis.

Lemma 35 (Affine-invertible row and column tuples). Let v be either the full variable tuple of one row or a nonempty subset of variables from one column. Suppose a block has input vector

$$g = M(y)v,$$

where M and a polynomial left inverse N have affine entries and $NM = I$ as an exact identity. Set $b^\top = \mathbf{1}^\top N$ and specialize the first extension factor's coefficients to b , all others to zero. This lifts every degree- D design for the Boolean linear-row base system to degree D , preserving old moments. Arbitrarily many such blocks can be processed through arbitrarily many levels with no degree inflation.

Proof. The selected sum is $\sum_a b_a g_a = \mathbf{1}^\top N M v = \sum_j v_j$. Thus each companion specializes to $g_a(1 - \sum_j v_j)$. For a row source this is $-g_a(\rho_i - 1)$, a degree-at-most-three base certificate. For a column source, set $s = \sum_j v_j$ and use

$$v_j(1 - s) = -(v_j^2 - v_j) - \sum_{k \neq j} v_j v_k.$$

Multiplication by the affine entries of M expresses $g_a(1 - s)$ by Boolean and column-exclusion axioms within degree three.

Every nonzero input has degree at least one and at most two, and its original companion degree is at least three. An allowed cofactor of degree $D - e_a$ therefore multiplies these certificates within degree D . The coefficients b are affine, so substitution does not increase degree and preserves old moments. For an affine $b = c + \sum a_j y_j$,

$$b^p - b = \sum_j a_j (y_j^p - y_j).$$

For a Boolean old variable use $y^p - y = (y^2 - y)(1 + y + \dots + y^{p-2})$; for an old extension variable use its field axiom. These are degree- p certificates, matching the original new field-axiom degree. The argument applies at each level, and compositions of affine substitutions remain affine. $\square$

Corollary 36 (An explicit globally coupled quadratic subclass). Take $M = I + A$ with affine entries and $A^2 = 0$, so $N = I - A$. These matrices satisfy Lemma 35, even when their entries involve every old variable. In particular, for an arbitrary affine $L(y)$ the row tuple

$$g_1 = x_{i1} + Lx_{i2}, \quad g_2 = x_{i2}, \quad g_j = x_{ij} \quad (j \geq 3)$$

has the first-factor coefficients $1, 1 - L, 1, \dots, 1$ and a degree-preserving lifting.

Proof. $(I - A)(I + A) = I - A^2 = I$. To ensure $A^2 = 0$, split the index set into disjoint parts I, J and allow nonzero entries only in positions $I \times J$. The displayed row tuple has sum $g_1 + (1 - L)g_2 + \sum_{j \geq 3} g_j = \rho_i$. Apply the preceding lemma. The nonlinearity is genuine when L is nonconstant; no row-exclusion axiom has been introduced. $\square$

Obstruction 37 (Even base-aware polynomial selectors can need large degree). Assume the imported residual PHP design bound. Let $2q \leq n$ and consider one block with inputs $g_a = x_{a,2a-1}$, $1 \leq a \leq q$. Suppose substitutions of maximum degree T make each specialized companion a consequence of the full base system $\mathcal{F}_n$, with certificate degree at most C . If $C \leq \lfloor (n - q)/2 \rfloor$, then

$$q \leq h(T + 1).$$

This is a limitation of an unconditioned polynomial substitution with short base certificates, not a limitation of every joint design.

Proof. For each $a \leq q$, assign pigeon a to one of holes $2a - 1, 2a$, encoding the choice by a Boolean parameter t_a : $x_{a,2a-1} = t_a$, $x_{a,2a} = 1 - t_a$. Remove the occupied hole and assigned pigeon. Use common residual variables z for all choices. For remaining pigeons, represent the paired columns by

$$x_{i,2a-1} = (1 - t_a)z_{i,a}, \quad x_{i,2a} = t_az_{i,a}.$$

For each fixed Boolean t , this is an ordinary partial-matching restriction to PHP with $n - q$ holes. Choose one normalized degree- C design μ for that common residual system.

Let $P(x) = \prod_u (1 - \sum_a \beta_{u,a}(x)x_{a,2a-1})$ and set $R(t) = \mu(P(x(t, z)))$, applying μ coefficientwise in the residual variables. The certificate-degree assumption implies $\deg P \leq C - 1$: P is nonzero because it equals 1 when all selected cells are zero, and $\deg(x_{a,2a-1}P) \leq C$ in the polynomial ring. Thus μ is defined on all residual coefficients. Each original variable has degree at most one in t , so $\deg_t R \leq h(T + 1)$.

At every Boolean t , applying the residual design to the restricted certificates gives $t_a R(t) = 0$ for all a . Also $R(0) = 1$. Its unique multilinear representative is therefore $\prod_a (1 - t_a)$, which has degree q over every field. Multilinear reduction cannot increase degree, yielding the inequality. $\square$

For $q = \Theta(n)$ and $h = O(\log n)$, the selector degree is $\Omega(n/\log n)$ in this model. A scalar conditioning that fixes one selected cell to 1 can handle this one block cheaply; it changes old moments and does not contradict the obstruction.

7 Joint moments and supported coefficient corrections

This section preserves the direct-functional route, even though later PC elimination has better global cost. Its purpose is different: several results extend a *prescribed* base functional without changing its low-degree old moments. Component functionals are unnormalized and may be nonzero while taking value zero on 1.

For one block, fix input index j and define

$$T_j(P) = [t_1 \cdots t_h] P(y, r_{u,j} = t_u, r_{u,k} = 0 \ (k \neq j)),$$

where $t_u^2 = t_u$ is imposed before extracting the coefficient. For a whole level, $T_{a,j}$ also sets new variables from other blocks to zero. Then $T_{a,j}(f(y)) = 0$ and $\deg_y T_{a,j}(P) \leq \deg P - h$.

Lemma 38 (Degree-window lifting over $\mathbb{F}_2$). Let the old system contain Boolean equations for every old variable. Consider an arbitrary level of nonlinear blocks of common accuracy h . If every active nonzero companion satisfies $D - h < e_{a,i} \leq D$, every degree- D old design extends to a degree- D design of the whole level, preserving all old moments.

Proof. Put $M = D - h$. For block a , let

$$V_a = \left\{ G = \sum_i q_i g_{a,i} : \deg q_i \leq D - e_{a,i}, \ i \text{ active} \right\}.$$

Choose unnormalized functionals $\mu_{a,j} : k[y]_{\leq M} \rightarrow k$ annihilating $\mathcal{I}_M(\mathcal{F})$ and satisfying

$$\sum_j \mu_{a,j}(G g_{a,j}) = \lambda(G) \quad (G \in V_a).$$

These products fit within M : with $\delta_a = \max_j \deg g_{a,j}$, $\deg G \leq D - h(\delta_a + 1)$.

Their consistency is a finite-dimensional dual condition. If $G g_{a,j} \in \mathcal{I}_M(\mathcal{F})$ for all j , write $G = \sum_i q_i g_{a,i}$ and set $k_a = \max_i (D - e_{a,i}) < h$. Then $G^2 = \sum_i q_i (G g_{a,i}) \in \mathcal{I}_{M+k_a}(\mathcal{F})$. Boolean

reduction of $G^2 - G$ stays within the same degree, so $G \in \mathcal{I}_{M+k_a}(\mathcal{F})$ and $\lambda(G) = 0$, since $M + k_a < D$. This proves that the map $G \mapsto ([Gg_{a,j}])_j$ has kernel annihilated by λ , and linear extension supplies all components.

Define

$$\Lambda(P) = \lambda(P(y, 0)) + \sum_{a,j} \mu_{a,j}(T_{a,j}(P)).$$

For a monomial cofactor H of $E_{a,i}$, the selected substitution is zero or $q(y)t_A$, where $|A| \leq \deg H < h$. Its full coefficient is

$$qg_{a,i}(1 - g_{a,j})^{|A|}(-g_{a,j})^{h-|A|}.$$

When $A = \emptyset$, this reduces to $qg_{a,i}g_{a,j}$ over $\mathbb{F}_2$. When $0 < |A| < h$, it is zero modulo Booleanity. The reductions stay within M . Corrections from another block vanish because H cannot provide one variable from all of that block's h groups. Thus the component equation cancels the base term. Old-axiom multiples are annihilated because $T_{a,j}(Hf) = fT_{a,j}(H)$; new Boolean equations reduce to zero in the auxiliary variables. Normalization and old moments are unchanged. $\square$

Corollary 39 (The cofactor-sensitive version). Set $k = \max_{a,i \text{ active}}(D - e_{a,i})$ and $B = \max\{D, D - h + k\} \leq 2D$. A degree- B base design supplies the same construction annihilating every allowed extension multiple whose cofactor monomials each miss at least one factor group in every block. Consequently, if a degree- $2D$ base design exists, every degree- D augmented refutation has an extension cofactor monomial touching all h groups of some block.

Proof. The consistency argument now uses $M + k \leq B$. The coefficient calculation and cross-block vanishing still hold under the stated support condition. A refutation using only such cofactors would be annihilated term by term while $\Lambda(1) = 1$, a contradiction. This is a necessary cofactor pattern, not a size lower bound. $\square$

Lemma 40 (The earlier odd-characteristic degree-window budget). Let input degrees be at most δ , and put $k = \max_{a,i \text{ active}}(D - e_{a,i})$. Over $\mathbb{F}_p$, a sufficient input degree for the analogous cofactor-sensitive construction is

$$B_p = D + \max\{0, (p-1)k + (2p-3)\delta - h\}.$$

If $k < h$, it gives a complete lifting for the level. The sharper binary Lemma 38 needs no additional δ term.

Proof. Use the operators

$$T_{a,j,\alpha}(P) = \chi_\alpha(g_{a,j})[t_1 \cdots t_h]P(y, r_{a,u,j} = t_u/\alpha, r_{\text{other}} = 0), \quad \alpha \neq 0,$$

in component domains $M = D - h + (p-1)\delta$. Couple the components by $\sum_{j,\alpha} \mu_{a,j,\alpha}(G\chi_\alpha(g_{a,j})) = \lambda(G)$. A dual kernel condition and the indicator identity imply $Gg_{a,j} \in \mathcal{I}_M(\mathcal{F})$. Write $G = \sum_i q_i g_{a,i}$ and use

$$G^p = \sum_i q_i G^{p-2}(Gg_{a,i}).$$

The total bounded-consequence degree is at most $M + k + (p-2)(k + \delta) = D - h + (p-1)k + (2p-3)\delta$. Reduction of $G^p - G$ uses no larger degree. Linear extension gives the components. On the support of $\chi_\alpha(g)$, the selected value of g is α , so nonempty proper factor-group supports vanish; full supports from other blocks are absent under the cofactor condition. Weight the sum of corrections by $(-1)^{h+1}$ to cancel the old-only coefficient. Old and new field equations are handled by the same finite-domain reductions. If $k < h$, every allowed multiplier has the required support restriction. $\square$

Obstruction 41 (Independent single-block corrections are not enough). With two constant-input blocks $g = 1$ and accuracy $h = 2$ over $\mathbb{F}_2$, a sum of single-block corrections can satisfy the individual blocks but fail on a cross-block multiple.

Proof. Write $R_A = r_{A,1}r_{A,2}$ and $R_B = r_{B,1}r_{B,2}$. Such a sum can have $\Lambda(R_A) = \Lambda(R_B) = 1$ but $\Lambda(R_AR_B) = 0$ and all intervening partial-group moments zero. Then

$$\Lambda(R_A(1 - r_{B,1})(1 - r_{B,2})) = 1.$$

Assigning the mixed moment $\Lambda(R_AR_B) = 1$ repairs this instance. This proves failure of the uncorrected construction, not nonexistence of a joint functional. $\square$

Proposition 42 (A sufficient mixed-component hierarchy). Over $\mathbb{F}_2$, index components by sets S of blocks and a selected input $J = (j_a : a \in S)$. Let $T_{S,J}$ extract the product of all $h|S|$ auxiliary variables from those selected groups, setting other new variables to zero. Suppose $\mu_{S,J}$ annihilates $\mathcal{I}_{D-h|S|}(\mathcal{F})$, with $\mu_\emptyset = \lambda$, and satisfies

$$\mu_{S,J}(qg_{b,i}) = \sum_j \mu_{S \cup \{b\}, J \cup \{b \rightarrow j\}}(qg_{b,i}g_{b,j}) \quad (b \notin S),$$

$$\mu_{S,J}(qg_{b,i}(1 - g_{b,j})) = 0 \quad (b \in S),$$

whenever $\deg q \leq D - e_{b,i} - h|S|$. Then $\Lambda(P) = \sum_{S,J} \mu_{S,J}(T_{S,J}(P))$ is a degree- D joint design.

Proof. For a cofactor monomial, a selected operator is zero if it uses incompatible input indices, if a selected unrelated block is missing a group, or if an unselected block variable remains. A nonempty proper support within the axiom's own block vanishes by Booleanity. The remaining cases are full group supports: adding the axiom's block produces the first displayed cancellation, and an already selected axiom block produces the second equation. Domains fit because $e_{b,i} = \deg g_{b,i} + h(\delta_b + 1)$ pays for the extra input factor. Old-axiom and field-equation verifications are the preceding coefficient calculations. Only the empty component contributes to 1. $\square$

Status. The conditions are sufficient. Their universal simultaneous feasibility was not proved. Later constructions replace this restricted ansatz; they do not retroactively establish every instance of this hierarchy.

Lemma 43 (Supported one-block moment lifting over $\mathbb{F}_2$). For one arbitrary nonzero-input block, put $\delta = \max_i \deg g_i$ and

$$\Phi_\delta(D) = \max\{D, 2D - h(\delta + 2)\}.$$

Every degree- $\Phi_\delta(D)$ old design extends, with all old moments through D preserved, to a degree- D design for the entire block and its field equations. No cofactor support restriction is needed.

Proof. Let $a = h(\delta + 1)$, retain only active indices i in the component equations, and set $N_i = D - h + \delta_i$. Let $V = \{G = \sum_i q_i g_i : \deg q_i \leq D - a - \delta_i\}$. Seek unnormalized ν_i annihilating $\mathcal{I}_{N_i}(\mathcal{F})$ and obeying

$$\sum_i \nu_i(Gg_i) = \lambda(G) \quad (G \in V).$$

If $Gg_i \in \mathcal{I}_{N_i}(\mathcal{F})$ for every active i , then

$$G^2 = \sum_i q_i(Gg_i) \in \mathcal{I}_{B_0}(\mathcal{F}), \quad B_0 = 2D - a - h = 2D - h(\delta + 2).$$

Indeed $N_i + \deg q_i \leq B_0$. Also $\deg(G^2 - G) \leq 2(D - a) \leq B_0$, so Boolean reduction gives $G \in \mathcal{I}_{B_0}(\mathcal{F})$. The base design annihilates G ; this is exactly the kernel condition for linear extension, proving existence of the components. If there are no active indices, take the zero specialization directly.

Now define

$$\Lambda(P) = \lambda(P(y, 0)) + \sum_i \nu_i(g_i T_i(P)).$$

The additional support factor g_i is the change from the earlier formula. For a selected cofactor $q(y)t_A$, the correction to HE_j is

$$qg_j g_i (1 - g_i)^{|A|} g_i^{h-|A|}.$$

For every nonempty A , including $A = [h]$, this is a Boolean-ideal polynomial because it contains $g_i(1 - g_i)$. For $A = \emptyset$, it reduces to $qg_j g_i$ and is cancelled by the component equation. All these expressions have degree at most N_i , as extracting the full auxiliary coefficient removes at least h degrees and the support factor adds δ_i . Incompatible selected indices give zero. Old-axiom multiples are annihilated since $g_i T_i(Hf) = f g_i T_i(H)$, and new Boolean equations reduce to zero. The corrections vanish on old polynomials and on 1. $\square$

Lemma 44 (Supported lifting over every fixed prime). With old Boolean or $\mathbb{F}_p$ -field equations, the analogous one-block sufficient budget is

$$\Phi_{p,\delta}(D) = \max\{D, pD - ph - ((p-1)h - (p-2))\delta\}.$$

It is at most pD and specializes to Lemma 43 for $p = 2$.

Proof. Replace $g_j T_j(P)$ by $\chi_\alpha(g_j)[t_1 \cdots t_h]P(r_{u,j} = t_u/\alpha, r_{\text{other}} = 0)$, with components $\nu_{j,\alpha}$ defined through $N_j = D - h + (p-1)\delta_j$. Couple them by $\sum_{j,\alpha} \nu_{j,\alpha}(G\chi_\alpha(g_j)) = \lambda(G)$ and multiply the correction sum by $(-1)^{h+1}$. A dual kernel condition gives $Gg_j \in \mathcal{I}_{N_j}(\mathcal{F})$. Using $G^p = \sum_j q_j G^{p-2}(Gg_j)$ gives degree at most

$$pD - (p-1)h(\delta+1) - h + (p-2)\delta = pD - ph - ((p-1)h - (p-2))\delta.$$

The same ceiling covers reduction of $G^p - G$. Thus the coupling equations are consistent. On the support of $\chi_\alpha(g_j)$, the selected input equals α ; every nonempty support has the factor $1 - g_j/\alpha$ and vanishes, including full supports. Auxiliary t variables are Boolean, so the new field equation maps to a scalar multiple of $t^p - t$ and reduces to zero. The remaining verifications are those of Lemma 43. $\square$

Corollary 45 (Two blocks and the first interaction boundary). Two blocks lift in order with sufficient input degree $\Phi_{p,\delta_A}(\Phi_{p,\delta_B}(D)) \leq p^2 D$. Over $\mathbb{F}_2$, if both have input degree δ and $D = h(\delta+2) + \delta$, the bound is $D + 3\delta$. For s equal-degree blocks with $D > C := h(\delta+2)$, repeated binary lifting has the sufficient budget $C + 2^s(D - C)$.

Proof. After the first lift, include its entire system and new field equations among the old axioms for the second. The second consistency argument then enforces cross-block conditions rather than assuming independent solutions agree. The formulas follow by composing $d \mapsto \max\{d, 2d - C\}$. At $D = C + \delta$ two iterations give $C + 4\delta = D + 3\delta$. Later inputs may use earlier variables if degrees are measured in that old-variable ring. $\square$

The exponential expression is a sufficient cost of this moment-preserving construction, not a proved lower bound on all lifting costs. Subsequent PC elimination avoids it by allowing the final base moments to change.

8 Why ordinary designs do not batch universally; the PC invariant

The next obstruction is not a PHP refutation. It disproves a proposed lifting principle for *arbitrary* Boolean base systems and explains why the program moved from ordinary designs to polynomial-calculus consequences.

Lemma 46 (A prefix-block refutation of degree $4h + 2$). Let a directed acyclic graph have vertices $1, \dots, N$ in topological order, a unique sink N , and at most two predecessors at each vertex. Set

$$b_i = 1 - x_i, \quad f_i = b_i \prod_{j \in \text{pred}(i)} x_j, \quad \mathcal{F}_G = \{f_i\}_i \cup \{x_N\} \cup \{x_i^2 - x_i\}_i.$$

For every prefix $[k]$ introduce a fresh one-level block on inputs $b_1, \dots, b_k$, with common accuracy h . Write

$$A_k = \prod_{u=1}^h \left(1 - \sum_{j \leq k} r_{k,u,j} b_j \right), \quad E_{k,j} = b_j A_k, \quad A_0 = 1.$$

The augmented system has a Nullstellensatz refutation of degree at most $4h + 2$, over every field, using no Boolean or new field axioms. Each certificate summand uses extension variables from at most two adjacent blocks.

Proof. Define

$$U_{k,j} = \sum_{u=1}^h r_{k,u,j} \prod_{v < u} \left(1 - \sum_{\ell \leq k} r_{k,v,\ell} b_\ell \right).$$

Telescoping gives $1 - A_k = \sum_{j \leq k} U_{k,j} b_j$, with $\deg U_{k,j} \leq 2h - 1$. Therefore

$$\begin{aligned} A_{k-1} - A_k &= A_{k-1}(1 - A_k) - A_k(1 - A_{k-1}) \\ &= \sum_{j < k} U_{k,j} E_{k-1,j} + U_{k,k} b_k A_{k-1} - \sum_{j < k} U_{k-1,j} E_{k,j}. \end{aligned}$$

Order the predecessors of k as $p_1, \dots, p_t$, $t \leq 2$. Then

$$b_k A_{k-1} = A_{k-1} f_k + b_k \sum_{\ell=1}^t \left(\prod_{v < \ell} x_{p_v} \right) E_{k-1,p_\ell}.$$

This follows from $1 - \prod_{\ell} x_{p_\ell} = \sum_{\ell} b_{p_\ell} \prod_{v < \ell} x_{p_v}$. Substitute this identity into each difference. The largest summand degree is $(2h - 1) + 2h + 3 = 4h + 2$; all others are no larger.

Sum the differences to obtain $1 - A_N$. Finally $A_N = A_N x_N + E_{N,N}$. Thus

$$1 = \sum_{k=1}^N (A_{k-1} - A_k) + A_N x_N + E_{N,N}$$

becomes the claimed bounded-degree certificate after the displayed expansions. Every term involves only block $k - 1$, block k , and original variables. $\square$

Obstruction 47 (No universal polylogarithmic batching from ordinary designs). Conditional on the published pebbling/NS degree correspondence, there are Boolean systems with ordinary designs through polynomially large degree that lose all degree- $O(\log N)$ designs after adding polynomially many linear-input blocks in a single level. Consequently no universal theorem of the form

$$\text{degree-}(D + h + \log M)^c \text{ base design} \implies \text{degree-}D \text{ augmented design}$$

can hold for every Boolean base system, for a fixed c .

Proof. The imported correspondence identifies the NS degree of a pebbling contradiction with reversible pebbling space, independently of the field [4, Theorem 3.1]. The cited Carlson–Savage graph properties give single-sink bounded-indegree families of $N = \Theta(r^3)$ vertices with standard pebbling price at least $r+2$; reversible space is at least that large [4, Section 4.2]. Hence the base NS degree is $\Omega(N^{1/3})$, and Lemma 2 gives designs below this bound. Take $h = \Theta(\log N)$, $D = 4h + 2$, and use the $N(N+1)/2$ prefix companions in Lemma 46. The hypothesized polylogarithmic input degree is below the base lower bound for large N , but the augmented refutation has degree D , a contradiction. $\square$

This also invalidates an argument based only on the number of complete groups in a monomial or on a sparse interaction graph: the certificate above uses a path of adjacent pairs. It does not show that every exponential lifting cost is necessary, or that a PHP-specific theorem is false.

Lemma 48 (The prefix base is easy for PC). The system $\mathcal{F}_G$ above has a PC refutation of degree at most three.

Proof. Derive b_k in topological order using

$$b_k = f_k + b_k \sum_{\ell=1}^t b_{p_\ell} \prod_{v < \ell} x_{p_v}.$$

The predecessor polynomials b_{p_ℓ} are already derived, so Lemma 3 permits the displayed products through degree three. For a source the equality is just $b_k = f_k$. Finally $b_N + x_N = 1$. $\square$

The previous obstruction is therefore consistent with cheap elimination into PC: high NS degree does not imply high PC degree. This distinction is why the PC-annihilation invariant excludes that counterexample.

Lemma 49 (One block from a PC-annihilating base functional). For a block over $\mathbb{F}_p$ with input degree at most δ and accuracy h , put

$$\Psi_{p,\delta}(D) = \max\{D, D - h + (p-1)\delta, p(D - h(\delta+1))\}.$$

A normalized functional annihilating $\mathcal{C}_{\Psi_{p,\delta}(D)}(\mathcal{F})$ yields an *ordinary* degree- D design of the augmented system, preserving old moments. In characteristic two, no degree loss occurs if $\delta \leq h$ and $D \leq 2h(\delta+1)$.

Proof. Use the supported operators of Lemma 44, but require the components in degree $N_j = D - h + (p-1)\delta_j$ to annihilate $\mathcal{C}_{N_j}(\mathcal{F})$. In the dual kernel condition, every Gg_j is now a derived PC polynomial. Write $a = h(\delta+1)$ and $G = \sum_j q_j g_j$ with $\deg q_j \leq D - a - \delta_j$. Then $\deg G \leq D - a$ and

$$G^p = \sum_j q_j G^{p-2}(Gg_j).$$

By Lemma 3, one reuses the derived polynomials Gg_j and pays only for the actual resulting product degrees, at most $p(D - a)$, not for their flattened certificate degrees. Together with the existing degree- N_j derivations, this is within $\Psi_{p,\delta}(D)$. Reduction of $G^p - G$ uses no larger degree. Thus G is a PC consequence annihilated by the input functional, proving component consistency. The supported-coefficient verification gives the ordinary output design. The stated binary no-loss conditions make each of the three terms in Ψ at most D . $\square$

Status. This lemma did not prove that its output annihilates PC consequences of the *augmented* system. Iterating it as though that stronger property were preserved would be invalid. The next section removes this gap by transforming actual PC derivations and then applying duality.

9 Additive elimination and nested-span batches

These are refutation transformations. They prove existence of some suitable augmented functional by contraposition and duality, not extension of every preassigned functional. Proof length can grow; the resource controlled is degree.

Lemma 50 (Eliminating one block at additive cost). Let $\mathcal{G}$ contain the Boolean or field equations for all old variables. A block with fresh variables and maximum input degree δ can be eliminated from a degree- d PC refutation at degree cost at most $(p-1)\delta$:

$$\mathcal{G} \cup \mathcal{E}_a \cup \mathcal{R}_a \vdash_d^{\text{PC}} 1 \implies \mathcal{G} \vdash_{d+(p-1)\delta}^{\text{PC}} 1.$$

The new variables must occur nowhere in $\mathcal{G}$.

Proof. Keep the given refutation π . Fix an input g_j and $\alpha \neq 0$. Set $r_{1,j} = \alpha^{-1}$ and all other new variables zero. Multiply every specialized line of π by $\chi_\alpha(g_j)$. An extension-axiom line becomes

$$\chi_\alpha(g_j)E_i(y, \beta) = \alpha^{-1}g_i(g_j^p - g_j),$$

an old-field consequence. Its actual degree is at most $d + (p-1)\deg g_j$ whenever that axiom occurs in π , and Lemma 4 supplies a certificate in that same degree. Old axioms become their polynomial multiples. New field equations vanish. Linear combinations commute with this map; multiplication by an old variable remains such multiplication, while multiplication by a specialized new variable becomes scalar multiplication. All mapped lines remain within the same degree ceiling. The final line is $\chi_\alpha(g_j)$, now derived from $\mathcal{G}$.

Repeat for all $\alpha \neq 0$ and use $g_j = \sum_{\alpha \neq 0} \alpha \chi_\alpha(g_j)$ to derive every input g_j from $\mathcal{G}$, through degree $d + (p-1)\delta$. Finally replay the original proof with all new variables zero. Every extension axiom becomes g_i . Reuse its already obtained derivation, rather than multiplying that derivation through every subsequent line. The replay refutes $\mathcal{G}$ within the claimed ceiling. Unused companions cause no obligation to map an axiom above the proof's degree; deriving extra inputs is harmless but unnecessary. $\square$

Lemma 51 (Additive elimination through all levels). For a leveled system of blocks with maximum input degrees δ_a , a degree- D augmented PC refutation yields a base PC refutation of degree at most

$$\boxed{D + (p-1) \sum_a \delta_a.}$$

Consequently, if the base has no PC refutation at that degree, there exists a normalized degree- D functional annihilating *all* degree- D PC consequences of the augmented system.

Proof. Remove blocks in reverse level order. When a block is removed, later blocks have already been removed; its fresh variables occur only in its own axioms and field equations. Lemma 50 therefore applies with all remaining axioms included in $\mathcal{G}$. Its output is an actual PC refutation, so the same hypothesis is available at every next stage. Add the degree increments. Contraposition and Lemma 2 give the functional. $\square$

This is stronger global compatibility than the preceding direct-design iterations supplied. It does not yet make polynomially many charges affordable.

Lemma 52 (Equal-span batch elimination). Suppose blocks in one level have the same polynomial input span V , with a basis of maximum degree δ_V . The whole batch is eliminated at cost at most $(p-1)\delta_V$, independent of its number of blocks and fan-in. If $1 \in V$, the batch has a constant specialization costing no degree.

Proof. For a basis element f of V , write $f = \sum_i c_{a,i} g_{a,i}$ in each block. Simultaneously set its first-factor coefficients to $\alpha^{-1}c_{a,i}$ and all other coefficients zero. Every companion in the batch then has the common factor $1 - \alpha^{-1}f$. The proof transformation with weight $\chi_\alpha(f)$ derives f from the remaining system through degree $d + (p-1)\delta_V$. Derive a whole basis, hence all inputs. Replay with zero coefficients and reuse the derived inputs. If $1 \in V$, choose coefficients summing to 1 in every block; their first factors are identically zero, so ordinary scalar substitution removes the batch without selectors. $\square$

Lemma 53 (Nested-span batch elimination). Let blocks $1, \dots, s$ belong to one level, have fresh mutually disjoint new variables, and have polynomial input spans

$$V_1 \subseteq V_2 \subseteq \dots \subseteq V_s.$$

If δ bounds all input degrees, a degree- D augmented PC refutation can be transformed into a refutation of the old system of degree at most

$$B = D + (p-1)\delta.$$

The chain can be arbitrarily long.

Proof. Keep the original degree- D proof π fixed. Process blocks in the displayed order, maintaining that all earlier inputs have been derived from the old system within the one ceiling B .

To learn an input f of block a , express f as a constant-coefficient combination of the inputs of each block $b \geq a$. For $\alpha \neq 0$, specialize those blocks to have first factor $1 - \alpha^{-1}f$ and their other factor coefficients zero; set all earlier blocks' coefficients to zero. Transform every line P of the original proof to $\chi_\alpha(f)P(y, \beta)$. Current and later companions become old-field consequences by Lemma 5. Earlier companions become $\chi_\alpha(f)g_{b,i}$, obtainable by reusing the already derived input $g_{b,i}$ and then multiplying its final polynomial. Their actual product degree is at most $D + (p-1)\delta$, since an axiom used in the source proof has degree at most D . Old axioms, new field equations, and inference rules are handled as in Lemma 50. The mapped proof derives $\chi_\alpha(f)$; combine over α to derive f within B .

Learn all inputs of block a and continue. No degree cost accumulates because every learning step maps the original proof, not a previously enlarged proof. After all inputs are learned, replay π with every new variable zero and reuse the derived input lines. The result is an old-system refutation through B . $\square$

Lemma 54 (Nested spans modulo supplied consequences). Let $H \subseteq k[y]_{\leq \delta}$ be a vector space whose elements have supplied PC derivations from the old system through degree c . Suppose

$$(V_1 + H)/H \subseteq \dots \subseteq (V_s + H)/H.$$

The batch can be eliminated at degree

$$B_H = \max\{c, D + (p-1)\delta, (p+1)\delta\}.$$

If $c \leq D$ and $D \geq 2\delta$, this is $D + (p-1)\delta$.

Proof. The representation of a selected input now reads $f = \sum_i c_{b,i} g_{b,i} + u_b$, where $u_b \in H$. The weighted companion image is

$$\alpha^{-1} g_{b,i} (f^p - f) + \alpha^{-1} \chi_\alpha(f) g_{b,i} u_b.$$

The first term is an old-field consequence; the second is a multiple of an already derived polynomial u_b . Both have degree at most $(p+1)\delta$. Reuse the supplied derivation of u_b and apply Lemma 3. All other steps of Lemma 53 are unchanged. This is a vector-space quotient with bounded derivations supplied, not the unrestricted ideal quotient of an unsatisfiable system. $\square$

Corollary 55 (Global chain budget). Partition each level into chains, with maximum input degree δ_C in chain C . A degree- D refutation eliminates to degree at most

$$D + (p-1) \sum_C \delta_C.$$

The same holds for chains modulo supplied consequences when the hypotheses making Lemma 54's bound additive hold at each elimination.

Proof. Remove levels in reverse order and whole chains one at a time. Other surviving axioms are old axioms for the current chain. The substitutions eliminate only the selected chain's fresh variables; they do not alter the input polynomials of other surviving blocks. Add the chain increments. $\square$

For example, 1000 nested linear-input blocks over $\mathbb{F}_2$, with $D = 49$, cost 50, not the earlier per-block bound 1049. This is a bound on proof degree, not a statement that the transformation has polynomial proof length.

10 Core-and-residual batching for incomparable families

Literal chains can be expensive to find, and incomparable spaces need not be intrinsically expensive. The next argument uses nested *cores* and spends the extension factors on the remaining directions.

Lemma 56 (Core-and-residual batch elimination). For blocks in one level let V_a be their polynomial input spans. Suppose

$$U_1 \subseteq U_2 \subseteq \cdots \subseteq U_s, \quad U_a \subseteq V_a,$$

where each core has generators of degree at most γ . Let $v_{a,1}, \dots, v_{a,r_a} \in V_a$ represent a basis of V_a/U_a , with degrees at most η_a . Define

$$T_a = \max \left\{ 1, \eta_a \left(\left\lceil \frac{(p-1)r_a}{h_a} \right\rceil - 1 \right)_+ \right\}, \quad T = \max_a T_a.$$

Use $\eta_a = 0$ when $r_a = 0$, and $\gamma = 0$ when all cores are zero. A degree- D augmented PC refutation can be eliminated to degree at most

$$B = TD + (p-1)\gamma.$$

There is no block-count or core-dimension factor.

Proof. We first construct a residual substitution for each block. Every input has a constant-coefficient decomposition

$$g_{a,i} = u_{a,i} + \sum_\nu c_{i,\nu} v_{a,\nu}, \quad u_{a,i} \in U_a.$$

Put $Z_a = \prod_{\nu}(1 - v_{a,\nu}^{p-1})$. As in Lemma 25, distribute its $(p-1)r_a$ elementary linear-in- v factors among the h_a groups and telescope within each group. Because each $v_{a,\nu}$ belongs to V_a , its coefficients are expressible in the original input tuple. This yields a polynomial substitution of degree at most T_a with

$$E_{a,i}(y, \beta_a) = g_{a,i}Z_a, \quad (g_{a,i} - u_{a,i})Z_a \in J_{\text{fld}}.$$

Thus, once the core has been derived, the block's specialized companions can be absorbed using core polynomials and old-field equations.

Keep the original degree- D proof π fixed. Process cores in order. Suppose earlier cores and the specialized companions of earlier blocks have already been derived from the old system through degree B . To learn a core generator $f \in U_a$, express f in the input span of every $b \geq a$; this is possible because the cores are nested. Use residual polynomial substitutions for earlier blocks, and scalar first-factor specializations $1 - \alpha^{-1}f$ for current and later blocks. Call the combined substitution Φ . Transform every original line to $\chi_\alpha(f)\Phi(P)$.

Current and later companions become $\alpha^{-1}g_{b,i}(f^p - f)$, old-field consequences. Earlier companion images are already derived; reuse their final polynomials and multiply by $\chi_\alpha(f)$. A mapped source line has degree at most $TD + (p-1)\gamma = B$. Old axioms become polynomial multiples of old axioms. New field equations map to zero or to $\beta^p - \beta \in J_{\text{fld}}$; a used field axiom has original degree $p \leq D$, so its image, even after the selector weight, fits within B .

For multiplication inferences, an original nonzero predecessor has degree at most $D-1$. Multiplication by a substituted variable is multiplication by a polynomial of degree at most T , implemented monomial by monomial. Every intermediate product stays within $T(D-1) + (p-1)\gamma + T = B$. Linear combinations commute with the map. Degree- nonincreasing old-field reduction justifies each field-ideal axiom image. The final line is $\chi_\alpha(f)$; combine over nonzero α to derive f within B . Repeating learns the whole core without charging its dimension.

Now absorb the current block. Its specialized companion is $u_{a,i}Z_a + (g_{a,i} - u_{a,i})Z_a$. If $Z_a = 0$, it is already zero. Otherwise, for a companion used in π , $\deg(g_{a,i}Z_a) \leq TD$, so $\deg Z_a \leq TD$. Since $u_{a,i}$ is a combination of core generators, its degree is at most γ . Hence both the core multiple and the old-field remainder have degree at most $TD + \gamma \leq B$. Reuse the derived core polynomial for the first and reduce the second by old-field equations. All used specialized companions of this block are now derived within B , completing the induction.

After all blocks, replay the original proof with their residual substitutions. Every axiom image is available, every inference fits the same ceiling, and the result is an old-system refutation. If all cores are zero, there are no learning steps and the argument is precisely simultaneous residual packing. $\square$

Corollary 57 (Inexpensive large antichains). For affine inputs, residual ranks $r_a \leq \lfloor 2h_a/(p-1) \rfloor$ give $T = 1$ and degree at most $D + p - 1$. For nonlinear residual inputs, $r_a \leq \lfloor h_a/(p-1) \rfloor$ gives $T = 1$ and degree at most $D + (p-1)\gamma$. The dimension of the nested cores is unrestricted.

Proof. In the affine case, each group has at most two elementary factors, yielding affine coefficients. In the nonlinear case, each group has at most one, yielding constant coefficients regardless of residual degree. Apply Lemma 56. $\square$

For example, over $\mathbb{F}_2$ let $V_a = U + P_a$ for a fixed large linear space U and different residual spaces with dimension at most $2h$ modulo U . The full spaces can be pairwise incomparable, yet the cores can all be U , so any number of blocks costs $D + 1$. At $h = 16$, $D = 49$, residual dimension 32 is affordable with output degree 50. This includes the first complete-factor-group cofactor regime.

Lemma 58 (Affine cores modulo supplied affine consequences). Let $H \subseteq k[y]_{\leq 1}$ consist of supplied old-system PC consequences through degree c . Form all input spaces in $k[y]_{\leq 1}/H$. If their images have nested cores and residual ranks as in Lemma 56, the affine batch has sufficient degree

$$B = \max\{c, TD + p - 1\} \quad (D \geq 2).$$

Proof. Choose affine representatives. A core relation is now $f = \sum_i c_{b,i} g_{b,i} + u_b$ with $u_b \in H$. The weighted future companion is

$$\alpha^{-1} g_{b,i} (f^p - f) + \alpha^{-1} \chi_\alpha(f) g_{b,i} u_b.$$

The first term is an old-field consequence and the second a multiple of a supplied consequence; both have degree at most $p + 1 \leq TD + p - 1$. Residual decompositions similarly acquire an affine H -term, whose supplied derivation is reused along with learned core polynomials in the absorption step. All other degree estimates of Lemma 56 are unchanged. $\square$

For PHP, the immediately available space is $H_0 = \text{span}\{\rho_i - 1\}$. Larger affine consequence spaces would have to be justified by bounded derivations; the rigidity result in the next section limits this possibility for base-only consequences.

Lemma 59 (Maximal cores for a proposed ordering). For an ordering $a_1, \dots, a_t$ of spaces, the largest possible nested cores are

$$U_{a_i} = \bigcap_{j=i}^t V_{a_j}.$$

They minimize each residual rank for that order.

Proof. The suffix intersections are increasing and belong to the corresponding spaces. If any other nested cores satisfy $U'_{a_i} \subseteq V_{a_i}$, then $U'_{a_i} \subseteq U'_{a_j} \subseteq V_{a_j}$ for every $j \geq i$. Thus U'_{a_i} lies in the suffix intersection. Larger cores have no larger quotient dimension. $\square$

Corollary 60 (Composing core-chain batches). Each batch gives a degree map $d \mapsto T_B d + (p-1)\gamma_B$. For batches with $T_B = 1$, the total charge is $(p-1)\sum_B \gamma_B$. A constant number of levels, each admitting a single batch with polylogarithmic T and γ , retains a polylogarithmic degree bound.

Proof. Apply the batch transformation in reverse level order and compose its affine degree maps. Scalar or old-variable substitutions remove only the current batch's fresh variables; surviving inputs are unchanged. For $T = 1$ the maps add, and a fixed number of polylogarithmic factors and summands remains polylogarithmic. $\square$

Status. No uniform inexpensive decomposition of all admissible input families was established. The next results solve the affine optimization inside this criterion and show that its best bound can exceed the PHP budget.

11 Exact affine decomposition and its sharp limitation

Assume common accuracy h and affine input spaces over $\mathbb{F}_p$, optionally in a quotient by supplied affine consequences. Remove zero spaces; spaces containing a nonzero constant can be handled separately by their zero-cost first-factor specialization. We optimize only the numerical bounds certified by the core-and-residual criterion. This is not optimization over all conceivable proof transformations.

Lemma 61 (Optimal ordering by dimension). For a batch S and an ordering $a_1, \dots, a_t$, set

$$R(a_1, \dots, a_t) = \max_i \left(\dim V_{a_i} - \dim \bigcap_{j=i}^t V_{a_j} \right).$$

An ordering of nondecreasing dimension minimizes this maximum. Its optimum $R_*(S)$ also equals

$$R_*(S) = \max_{\emptyset \neq A \subseteq S} \left(\min_{a \in A} \dim V_a - \dim \bigcap_{a \in A} V_a \right).$$

Proof. Suppose adjacent spaces A, B have $\dim A \geq \dim B$, and let W be the intersection of all spaces after them; for an empty suffix use the ambient space. Before swapping, the residual at A is $\dim A - \dim(A \cap B \cap W)$. After swapping, the affected residuals are $\dim B - \dim(A \cap B \cap W)$ and $\dim A - \dim(A \cap W)$, both at most that old residual. All other suffix intersections are unchanged. Repeatedly remove inversions to obtain the dimension order without increasing the objective.

For the second formula, take any nonempty subfamily A . The first member of A in an arbitrary order has a suffix containing all of A , so its residual is at least the bracketed expression. Conversely, in nondecreasing dimension order, every stage removes a smallest-dimensional space from the remaining family. Its residual is exactly the bracket for that remaining family. The maximum over all subfamilies bounds every stage, proving equality. $\square$

Lemma 62 (Optimal batch map and exact partition recursion). Put

$$T(r) = \max \left\{ 1, \left\lceil \frac{(p-1)r}{h} \right\rceil - 1 \right\}, \quad r_{\max}(S) = \max_{a \in S} \dim V_a.$$

For nonempty S , the best bound supplied by our affine core criterion is

$$C_S(d) = \min \{ T(R_*(S))d + (p-1), T(r_{\max}(S))d \}.$$

The best bound over partitions and batch order in one level is exactly

$$B(\emptyset) = D, \quad B(S) = \min_{\emptyset \neq A \subseteq S} C_A(B(S \setminus A)).$$

Proof. A nonzero affine core costs $p-1$ in the criterion; maximal suffix cores minimize all residual ranks, and Lemma 61 optimizes their order. If every core is zero, there is no core-learning charge but the largest residual is $r_{\max}(S)$. Intermediate cores cannot improve either case: a nonzero core still has degree one, and making it smaller only increases residual ranks. This gives the two alternatives in C_S .

Each C_A is nondecreasing in its input degree. In an optimal sequence with last batch A , the best preceding degree is therefore $B(S \setminus A)$; a worse preceding degree cannot help. Minimize over the last batch to obtain the recurrence. It is an exact finite dynamic program, exponential in the number of blocks, not a proof that its value is affordable. $\square$

Lemma 63 (A large affine space untouched by the row equations). For PHP with $n+1$ rows, set

$$H_0 = \text{span}\{\rho_i - 1\}, \quad W = \text{span}\{x_{ij} : i \leq n+1, j < n\}.$$

Then $\dim W = n^2 - 1$, and $k \oplus W$ injects into $k[x]_{\leq 1}/H_0$.

Proof. There are $(n+1)(n-1) = n^2 - 1$ distinct coordinate variables. If $w = c + \sum_i a_i(\rho_i - 1)$ with $w \in W$, comparison of the absent last-column coefficients $x_{i,n}$ forces every $a_i = 0$. Then a homogeneous linear w cannot equal a nonzero constant. This proves the asserted injection. $\square$

Lemma 64 (Many large pairwise disjoint input spaces). Fix p and sufficiently large n . Let

$$d_0 = \lceil \log_p n \rceil, \quad q = \left\lfloor \frac{n^2 - 1}{2d_0} \right\rfloor, \quad r = qd_0.$$

There are n subspaces of W , each of dimension $r = \Theta(n^2)$, whose pairwise intersections are zero both literally and modulo H_0 .

Proof. Let $K = \mathbb{F}_{p^{d_0}}$, which has at least n elements. Embed the $2r$ -dimensional $\mathbb{F}_p$ -space $K^q \oplus K^q$ into W . For n distinct $\alpha \in K$, use the graph spaces

$$V_\alpha = \{(u, \alpha u) : u \in K^q\}.$$

Each has dimension r . Equality $(u, \alpha u) = (v, \beta v)$ for $\alpha \neq \beta$ forces $u = v$ and $(\alpha - \beta)u = 0$, hence $u = 0$. The injection in Lemma 63 preserves these intersections in the quotient. The floor loses only $O(\log n)$ from $(n^2 - 1)/2$, giving the asymptotic dimension. $\square$

Obstruction 65 (The exact optimum of the static criterion can exceed the budget). Use a basis of each V_α in Lemma 64 as an input tuple for one fresh level of accuracy- h ENS blocks. There are n blocks, $nr = O(n^3)$ companions, and all inputs are homogeneous linear. For this family the best bound of Lemma 62 is

$$B_{\text{criterion}}^{\text{opt}} = \min\{D + (p - 1)n, T_r D\}, \quad T_r = \max\{1, \lceil (p - 1)r/h \rceil - 1\}.$$

For fixed p , $h = \Theta(\log n)$, and $2h + 1 \leq D = \text{polylog } n$, it equals $D + (p - 1)n > n/2$ for large n .

Proof. In any batch of at least two graph spaces, the first core lies in the intersection with a later space and is zero. Its residual already has dimension r , so $R_*(S) = r$. An empty-core batch costs $T_r d$ and the nonzero-core alternative is no better. A singleton costs $\min\{d + p - 1, T_r d\}$. Any partition using a multiblock batch or an empty-core singleton has total bound at least $T_r D$, since all maps are nondecreasing and do not decrease degree. Otherwise all n blocks pay the singleton core charge, giving $D + (p - 1)n$. Both alternatives are attained, proving the exact optimum.

Here $r = \Theta(n^2)$ and $D \geq 2h + 1$, so $T_r D = \Omega(n^2)$, while $D + (p - 1)n = \Theta(n)$ for fixed polylogarithmic D . The blocks are syntactically admissible, all companions are present, and their original degrees are $2h + 1$. The construction does not assert that the augmented PHP system has a refutation. $\square$

At $p = 2$, $n = 256$, $h = 8$, $D = 25$, one has $r = 32760$ and $T_r = 4094$. The optimum is $\min\{281, 102350\} = 281$, above the 128 target budget. This illustrates the criterion, not a matched Frege simulation instance.

Lemma 66 (Affine-consequence rigidity of the PHP base). Assume External input E1 for the base and its partial-matching restrictions. For $n \geq 4$ and $1 \leq b \leq \lfloor (n - 2)/2 \rfloor$,

$$\mathcal{C}_b(\mathcal{F}_n) \cap k[x]_{\leq 1} = H_0.$$

Proof. The inclusion of H_0 is immediate. Suppose $f = a_0 + \sum_{i,j} a_{ij} x_{ij}$ has a degree- b derivation. Choose distinct rows i, i' and columns j, k . Restrict by the two partial matchings $i \mapsto j, i' \mapsto k$ and $i \mapsto k, i' \mapsto j$. Both leave the same residual PHP system with $n - 2$ holes; restriction of a PC derivation does not increase degree. The resulting affine polynomials differ by the constant $a_{ij} + a_{i'k} - a_{ik} - a_{i'j}$. A nonzero constant would give a degree- b residual refutation, contrary to E1. All rectangle differences are therefore zero, implying $a_{ij} = u_i + v_j$.

Subtract row-axiom combinations to derive $f' = c + \sum_j v_j \sigma_j$, where $\sigma_j = \sum_i x_{ij}$. If $v_j \neq v_k$, permute these columns in the proof and subtract; division by the nonzero difference derives $\sigma_j - \sigma_k$ at degree b . Match one pigeon to hole j . In the residual system this becomes $1 - \sigma'_k$. Permuting its remaining columns derives $1 - \sigma'_\ell$ for every residual hole. Summing these $n - 1$ equations and the n residual row equations gives

$$\sum_{\ell=1}^{n-1} (1 - \sigma'_\ell) + \sum_{i=1}^n (\rho'_i - 1) = -1,$$

another forbidden degree- b refutation. Hence all v_j are equal. The row equations then reduce f to a constant, which must be zero by E1 for the original base. Thus $f \in H_0$. $\square$

Corollary 67 (More low-degree affine base consequences do not repair the barrier). Within the degree range of Lemma 66, quotienting the input spaces of Lemma 64 by all available affine base-PC consequences creates no additional intersections beyond quotienting by H_0 . The static optimum in Obstruction 65 is unchanged.

Proof. The consequence space is exactly H_0 by rigidity, and the graph spaces inject modulo H_0 by Lemma 63. Thus all dimensions and intersections used by the optimization remain the same. $\square$

Scope. The counterexample concerns the certified cost of a static affine core-and-residual decomposition. It does not show a low-degree ENS refutation, essential use in a translated Frege proof, or a lower bound for every elimination method. Consequences involving surviving extension blocks, nonlinear methods, conditional functionals, and certificate-sensitive rewritings are not excluded. What is ruled out is the hope that a better ordering or more low-degree *affine base-only* consequences always make this same criterion affordable.

12 Current state and the remaining route to the payoff

12.1 The strongest usable checkpoint

The manuscript supplies a sequence of working transformations, not a completed Frege lower bound. The most direct global checkpoints are:

$$\begin{array}{ll} \text{Individual blocks:} & D + (p - 1) \sum_a \delta_a, \\ \text{Equal spans or nested chains:} & D + (p - 1) \sum_C \delta_C, \\ \text{A core-and-residual batch:} & TD + (p - 1) \gamma. \end{array}$$

These handle all original-joint-degree cofactor conditions, all new field equations, and level dependencies when applied in the stated order. They imply existence of an augmented PC-annihilating functional whenever their final base degree stays below the imported PC lower bound.

The static affine decomposition has now been optimized, and an admissible family has a proved unaffordable optimum *within that criterion*. Thus the next step is not another search for a better ordering of the same spaces.

Open target 68 (Refutation-sensitive, affordable global elimination). Fix p , a proof depth ℓ , and a size exponent K . For every degree- D NS certificate actually produced by the simulation of a hypothetical size- n^K ordinary-PHP proof, construct a base PC refutation of degree $B \leq \lfloor n/2 \rfloor$, for all sufficiently large n .

There is no proof of this target in the conversation or this manuscript. It is the highest-risk missing theorem. A sufficient, stronger quantitative estimate would be $B \leq (D + h + \log M)^{O_{p,\ell}(1)}$, but that particular expression is not necessary: any justified bound fitting under the available linear degree budget suffices.

12.2 Remaining obligations, riskiest first

1. Use the certificate, not just its static input spaces. Prove that relevant extensions can be pruned, rewritten, or jointly eliminated at affordable cost using their actual cofactors, proof dependencies, or PHP-specific information. The graph-space family shows that an affordable static decomposition cannot be assumed for every admissible family. A successful argument may exploit nonlinear consequences or relations involving surviving extensions, neither of which is addressed by affine base rigidity. It may instead construct a joint functional directly. These are possibilities, not proved reductions.

2. Establish coverage of the complete translated output. The published simulation supplies a size, degree, accuracy, and level regime; it does not assert favorable chains or cores. The new theorem must cover nonlinear inputs, dependencies on earlier levels, all companion polynomials, and new field equations. It is enough to handle the translated NS certificates. Our current lemmas also handle arbitrary PC refutations, but this stronger scope is not required for the payoff and may be an unnecessary obstacle.

3. Independently audit the working elimination lemmas. Check every transformed PC line, degree-bounded finite-domain reduction, original-degree convention, reuse of derived inputs, treatment of inactive companions, and reverse-level freshness. For consequence quotients, the relevant derivations must actually exist at the stated degree. The archived examples do not certify these general statements. Consolidation gives explicit proof obligations; it is not a substitute for independent verification.

4. Complete the ordinary-PHP encoding transfer. The required direction is

short ordinary-PHP refutation $\implies$ short refutation of the Boolean linear-row algebraic system.

Semantically a Boolean row with sum 1 modulo p is nonempty. The remaining proof-level task is a bounded-depth, polynomial-overhead derivation of that implication. Do not silently add row exclusion or identify the two encodings. The exact encoding in the imported PC lower bound must also be checked; deleting extra axioms is legitimate, replacing equations without a transfer is not.

5. Close every parameter quantifier. For every fixed p, ℓ, K , obtain the translated parameters and ensure the elimination theorem covers all their constants for sufficiently large n . The proof must not assume $p > n$, cover only a fixed number of blocks, or handle only one polynomial size exponent. Polynomially many companions are distinct from polynomially many blocks, and h is a factor count within a block.

Lemma 69 (The payoff implication, conditional on the missing obligations). Assume External inputs E1–E2, the ordinary-PHP encoding transfer just stated, and Open target 68. Then for each fixed prime p and every fixed proof depth, ordinary PHP has superpolynomial $\text{AC}^0[p]$ -Frege proof-size lower bounds.

Proof. Fix p, ℓ, K and suppose a size-at-most- n^K proof exists. Apply the encoding transfer and E2. For logarithmic accuracy the resulting NS degree is $D = \text{polylog } n$, with polynomially many companions and constantly many levels. Open target 68 gives a base PC refutation of degree at most $\lfloor n/2 \rfloor$. This contradicts E1. Since the contradiction holds for every fixed ℓ, K once n is large

enough, it gives

$$\forall \ell, K \exists n_0 \forall n \geq n_0 : \text{Size}_{\ell,p}(\text{PHP}_n) > n^K.$$

This is only a conditional implication; its central elimination hypothesis remains open. $\square$

12.3 What is no longer a separate requirement

The exact erosion formula, a sharper generic-linear survival threshold, and survival until complete dimension exhaustion are not necessary for the current PC-elimination route. The non-Boolean generic theorem is a separate algebraic result, not a completed transfer to Boolean ENS. A universal cheap lifting from ordinary base designs is false by the prefix construction. A universal cheap static affine decomposition is false within our criterion by the graph-space construction. Neither negative statement is a lower bound on all possible PHP-specific eliminations.

12.4 Compact correction and limitation register

Earlier shortcut or aspiration	Correct scope retained here
Koszul vanishing for all $d < n$	False; the proved working range is $2d - 1 \leq n$, with a characteristic-independent counterexample outside it.
Positive quotient dimension	Does not imply survival of the specified vector z^d ; normalized separation is a separate condition.
Generic linear survival is Lemma B for ENS	Not without a transfer handling Booleanity, nonlinear products, field equations, and proof-derived parameters.
Every scalar specialization should work	False even for the input $g = 1$. The required functional is existential for each extension instance.
Specialization lowers the cofactor threshold	Our bounds retain original joint-variable axiom degrees. Regrading can create unproved extra obligations.
Annihilate factors individually	A functional is not multiplicative. All allowed axiom multiples must be annihilated.
Condition once per block without rechecking	Later weights can spoil moment cancellation. The repair algorithm rechecks unhandled blocks.
Make all weighted companions vanish modulo field/column equations	Can force a degree above n ; row equations can bypass this pointwise obstruction.
One low-degree polynomial substitution handles every block	Diagonal inputs give a selector-degree obstruction even with bounded base certificates.
Independent local moment components glue	False without compatibility; supported corrections or proof transformations must supply the coupling.
Few blocks per monomial or a sparse interaction graph suffice	The prefix certificate has only adjacent-pair interactions yet defeats ordinary-design batching.
A PC-based input lemma automatically preserves PC annihilation	It did not; actual PC elimination and duality supply a valid global invariant.
A better static ordering will always fit the budget	The ordering problem is solved, and the static criterion has admissible families with an unaffordable optimum.
Archived tests prove the general theorems	They validate their stated finite instances or proof transformations only.

A Evidence ledger and reproducibility

No research tests were rerun for this compilation. The following figures are read from the previously generated archives and the conversation. They are included so that a future researcher can distinguish mathematical proofs from the actual scope of the experiments. Successful checks on satisfiable finite domains test identities and functionals there; they do not establish the existence of truncated designs for an unsatisfiable PHP instance. Checking a finite list of primitive PC proof lines is stronger evidence for an implemented transformation on those examples, but still not a proof of the transformation for every input.

The source bundle preserves the original ZIP archives without silently editing their results. It also contains a manifest of filenames, byte sizes, members, and SHA-256 checksums. No font files are included.

Archived runs, in research order

Construction	Recorded checks	What they establish in the tested instances
Generic linear functional	30 feasible cases; 4 infeasible controls; 3 determinant-chart checks	Exact finite-field checks in the original non-Boolean ring. No Boolean ENS claim.
Reweighted lifting	46 cases; 732 active companions; 26,988 multiplier tests	Scalar repair and moment rechecking. The original joint-variable degree budget is retained.
Factor packing	76 cases over four primes; 125,706 companion/assignment checks; 1,824 multiplier moments	Polynomial substitution identities and degree bounds; 36 independent-input controls attain the selector bound.
Shared conditioning	48 cases; 1,536 blocks through four levels	Common feature weights, recursive scalar choices, and preservation of specified old axioms on satisfiable finite domains. Not full PHP designs.
Base-aware affine matrices	18 cases; 576 blocks; 3,079,296 companion/assignment checks; 9,216 mixed moments	Affine inverse substitutions. Fifteen cases include row-normalized assignments with repeated row occupancy; row exclusion is not silently assumed.
Degree-window joint moments	42 cases; 126 blocks; 396 companions; 24,654 annihilation checks	Twelve cases have genuinely unsatisfiable Horn-chain bases with truncated designs. A boundary control detects failure without mixed components.
Supported two-block moments	23 binary cases; 30,063 cofactor tests; 476 explicit complete-group checks	The actual two-stage component construction. Four unsatisfiable-base cases use constant inputs; nonconstant cases use satisfiable bases. Odd-prime formulas were not tested here.

Construction	Recorded checks	What they establish in the tested instances
Prefix batching obstruction	16 formal expansions; 32 negative controls; 8,192 evaluation assignments	An exact degree- $4h + 2$ certificate and its degree accounting. Not the external pebbling lower bound or the PC-based lifting.
Nested-span elimination	14 cases; 4,500 source and 9,670 output proof lines	Primitive PC line checking, including 328 reuses of learned inputs. Synthetic propagation bases, not PHP. The consequence-quotient variant was not tested.
Core-and-residual elimination	7 cases; 6,440 source and 1,917 output proof lines; 26,298 mapped-line identities	Pairwise incomparable input spaces, scalar and affine residual substitutions, and nonlinear cases. All tested regimes have $T = 1$. Not the general quotient variant.
Affine decomposition	16 cases; 34,796 graph-space pairs; 17,604 orderings; 12 quotient-rank checks	Exact subspace calculations, exhaustive small-order comparisons, and small subset dynamic programs. Not a PHP/ENS refutation or computational proof of affine rigidity.

Archive filenames

1. annihilating_functional_checks.zip
2. ens_lifting_checks.zip
3. factor_packing_lift_checks.zip
4. shared_conditioning_packing_checks.zip
5. base_aware_lifting_checks.zip
6. joint_moment_lifting_checks.zip
7. two_block_lifting_checks.zip
8. batch_lifting_attack_checks.zip
9. nested_batch_elimination_checks.zip
10. core_residual_batch_elimination_checks.zip
11. decomposition_optimization_checks.zip

A separate last-turn identity-only check of the additive-elimination selector identities reported 28 exact checks over five primes; it was not a full proof-transformation test. Its quoted runtime and success count are not upgraded here into verification of Lemma 50.

The core-and-residual archive records an initial timed-out attempt and later successful optimized runs. That unsuccessful attempt is part of the historical performance record, not a passed test. Other archives retain their own timing scope, including overlapping parallel worker times that must not be summed as wall-clock time.

Computational standard used in the conversation

Finite-field calculations were intended to use exact modular arithmetic, vectorized array operations or compiled elimination, and independent parallel cases with numerical-library thread counts controlled to avoid oversubscription. Floating-point matrix rank is not a substitute for rank over

$\mathbb{F}_p$. The archives describe their actual implementations and limits. This manuscript makes no new performance or formal-verification claim about them.

B Source concordance and assembly notes

The authoritative source for the new mathematics in this document is the visible research conversation, not a claim that the original uploaded PDF already contained these arguments. This concordance identifies the main stages without reproducing the repeated route-planning messages.

Conversation stage	Consolidated location
Attempting item (a)	Section 2: false broad range, multigraded induction, truncated Hilbert function.
Finding the generic annihilating functional	Section 3: selected-column specialization, maximal rank, determinant chart.
Specifying admissible extension data	Definition 6, Lemma 7, and the Boolean target after Lemma 18.
Finding a nonlinear lifting	Lemmas 19–21: exact scalar repair and one-level reweighting.
Reducing its cost / packing factors	Lemmas 25–29: parallel factor packing and hybrid choices.
Packing shared inputs	Lemmas 22–24: shared features and whole-column conditioning.
Testing the proposed uniform packing step	Lemmas 31–33: pointwise obstruction and its row-axiom escape.
Using the base equations	Section 6: bounded certificates, affine-invertible tuples, and diagonal selector obstruction.
Constructing compatible joint moments	Section 7: degree windows, full-group failures, hierarchy, supported corrections.
Attacking block-count growth	Section 8: prefix certificate, ordinary-design obstruction, stronger PC input.
Additive PC elimination	Lemmas 50–52: actual refutation transformations and common-span batches.
Nested batch elimination	Lemmas 53–55: replay of one source proof, learned inputs, base consequences.
Attacking incomparable families	Section 10: nested cores and factor-packed residual directions.
Optimizing the decomposition	Section 11: exact ordering, partition recursion, graph-space counterexample, affine rigidity.
Remaining payoff route	Section 12: proof-sensitive elimination, coverage, audits, encoding, and quantifiers.

Assembly qualifications. Statements about degree are written in the ordinary polynomial ring before quotient reduction, unless a separate normal-form degree is explicitly defined. The field-reduction lemma supplies bounded certificates for finite-domain ideal membership. Repeated direct-design results are retained because they preserve old moments, whereas later PC elimination only guarantees existence of some compatible functional. Numerical observations unsupported by a surviving theorem appear as records or in the ledger, not as lemmas. The initial strong generic exhaustion claim and the unrestricted cheap decomposition aspirations remain withdrawn or open.

What is not claimed by the word “proof.” The included proofs are the working mathematical arguments from this research, organized and made explicit. They have not been formally verified, refereed, or independently audited. Imported lower bounds remain imported premises. The compiled result is a research handoff that makes verification possible, not a certification that the intended open payoff has been achieved.

Build and timing record. The companion source bundle contains timestamped phase boundaries and subprocess runtimes. Marked reading windows include interpretation. A source-lookup window includes network, server, tool-dispatch, and interleaved review time; these are not separable from this interface. Build, rendering, and local PDF checks are timed as subprocess wall-clock intervals. The remaining interval includes reasoning, drafting, coding, layout review, and unisolated tool overhead. Final response generation occurs after the last measurement and is not represented as timed. No mathematical test suite was executed during this PDF build.

References

- [1] S. Buss, R. Impagliazzo, J. Krajíček, P. Pudlák, A. Razborov, and J. Sgall. *Proof complexity in algebraic systems and bounded depth Frege systems with modular counting*. Computational Complexity 6 (1996/1997). doi:10.1007/BF01294258. The original simulation source; the form used here is exposed in [2].
- [2] Jan Krajíček. *Extended Nullstellensatz proof systems*. arXiv:2301.10617, version 3 (27 September 2023). arXiv:2301.10617. Definition 2.1 supplies the extension syntax; Lemma 4.1 records PHP designs; Theorem 5.2 records the Frege-to-ENS simulation. These are external inputs.
- [3] Alexander A. Razborov. *Lower bounds for the polynomial calculus*. Computational Complexity 7 (1998), 291–324. doi:10.1007/s000370050013. The imported linear degree lower bound for algebraic PHP; the encoding match for the final ordinary-PHP payoff must be checked explicitly.
- [4] Susanna F. de Rezende, Or Meir, Jakob Nordström, and Robert Robere. *Nullstellensatz Size-Degree Trade-offs from Reversible Pebbling*. arXiv:2001.02481 (2020); preliminary version, CCC 2019. arXiv:2001.02481. Theorem 3.1 and Section 4.2 are the imported premises in Obstruction 47.
- [5] The Stacks Project Authors. *The Koszul complex*, Tag 0621; and *Koszul regular sequences*, Tag 062D. Tag 0621, Tag 062D. Background for the Koszul identities used in Sections 2 and 3; the specialized arguments are given in the manuscript.
- [6] User-supplied historical starting record. *Full state of the investigation: The pigeonhole principle and $AC0[p]$ -Frege: designs, the BIKPRS route, and where it stands*. research-state-full.pdf, 9 pages. Listed for provenance only. The present manuscript records the subsequent conversation and does not certify or reproduce all claims in the starting PDF.
- [7] The research conversation compiled in this document, from the request to attempt item (a) through the affine decomposition optimization. The eleven generated check archives are listed in Appendix A; the source concordance is Appendix B.